

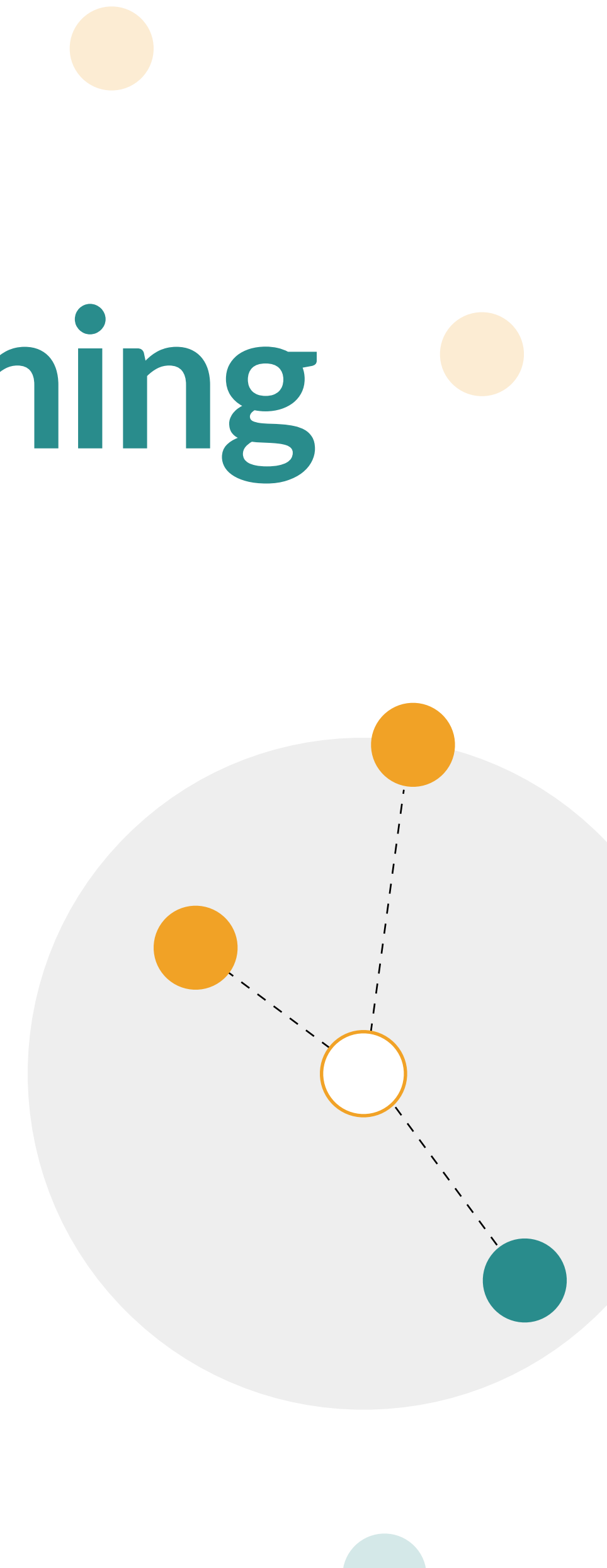
Foundations of Machine Learning

Vito Dichio, Fellow in AI (ENS-PSL)

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Lecture 1 : Introduction

Master's Degree in **AI & Society** (PSL-PSAI, Fall 2026-27)



AI Architecture M1@IASO

Supervised Learning

Linear Regression & Classification
Trees & Random Forests
Regularisation, Kernels, Basis
Bias-Variance & Model Selection
Support Vector Machines

Unsupervised Learning

Clustering
Principal Component Analysis
Graphical Models
EM & Hidden Markov Models

S1

Foundations of
Machine Learning

A theory-first foundations course
Minimal coding, no NNs



Learn to « **think like an ML scientist** »:

Ask why a method works
Diagnose failure—overfitting, wrong assumptions...
Choose the right tool for the data
Read a new method and place it in the landscape you already know

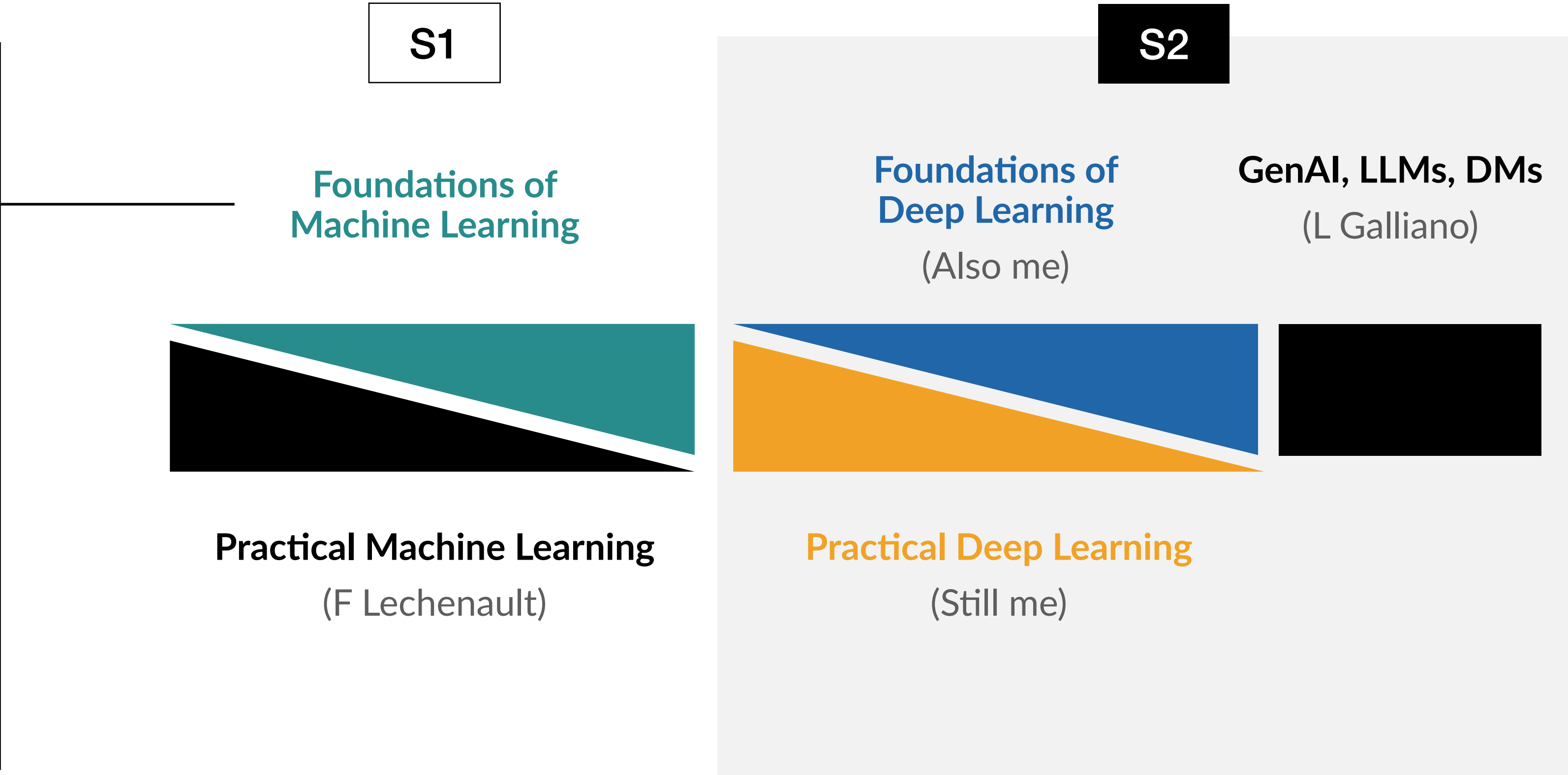
AI Architecture M1@IASO

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Course in Practice



Lectures /

Thursday@1:45PM (17/9 - 17/12)

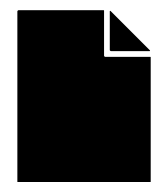
◆ *No lecture on 12/11 and 26/11*



Office hour /

Thursday@12:30PM

Didn't fully land in lecture? This is the place to clear it up—no question too small. Drop in, no need to book.



Course materials /

Lectures & scripts on  **moodle**
Running notes on  **CO** ([link here](#))

Course in Practice



Lectures /

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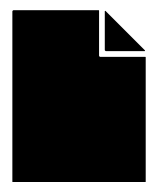
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EXAM

30-min oral

Format /

Chalk & blackboard, covering both Supervised and Unsupervised Learning

Support /

Relevant formulas provided when appropriate—this is NOT a memorisation test

What's assessed /

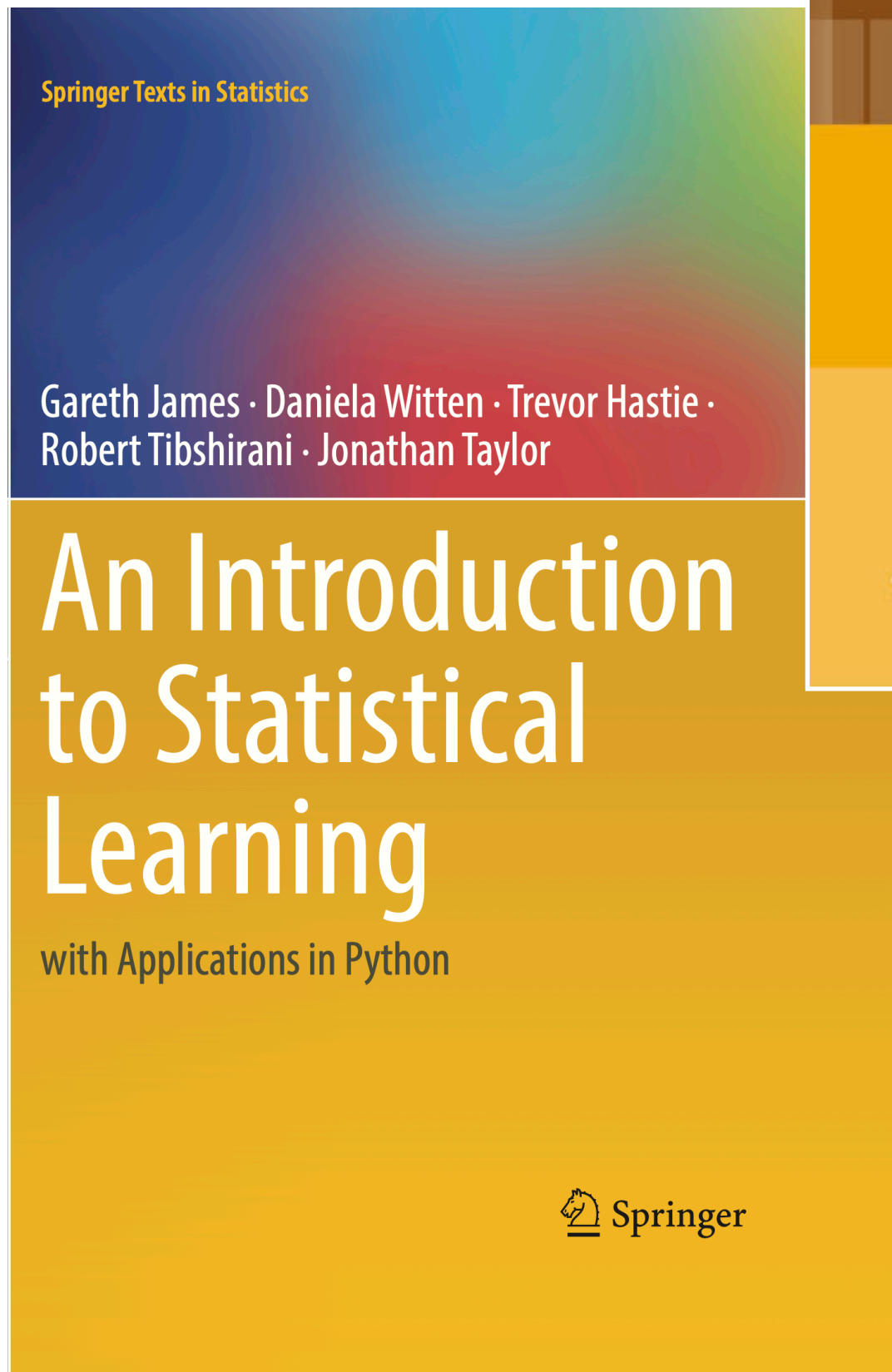
Your ability to reason through a method—explain why it works, derive key steps, and connect ideas across the course

Date /

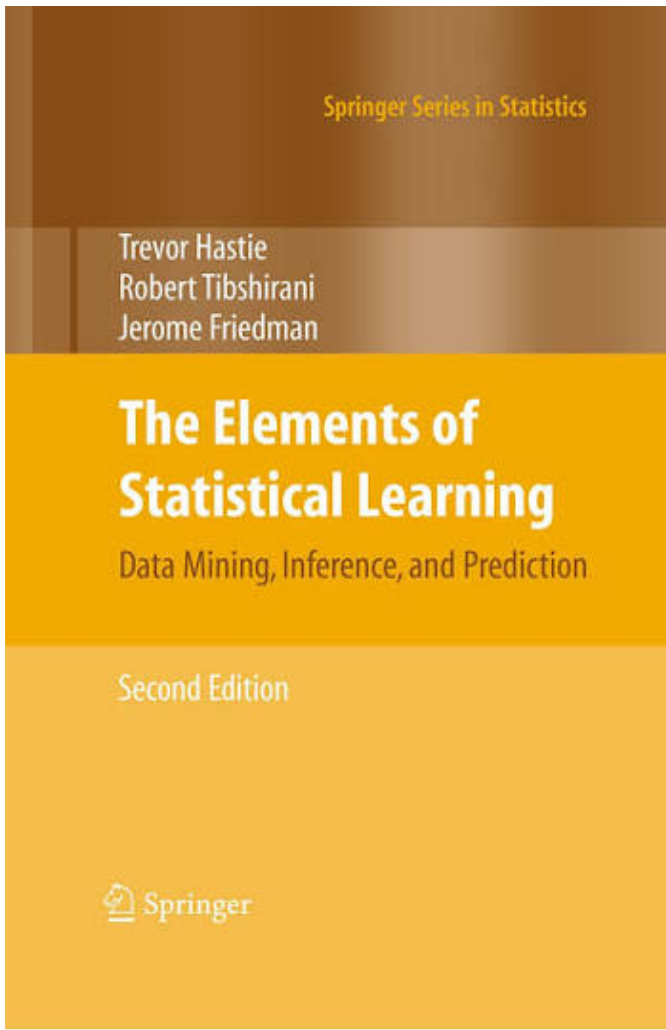
11-15/1/2027

References

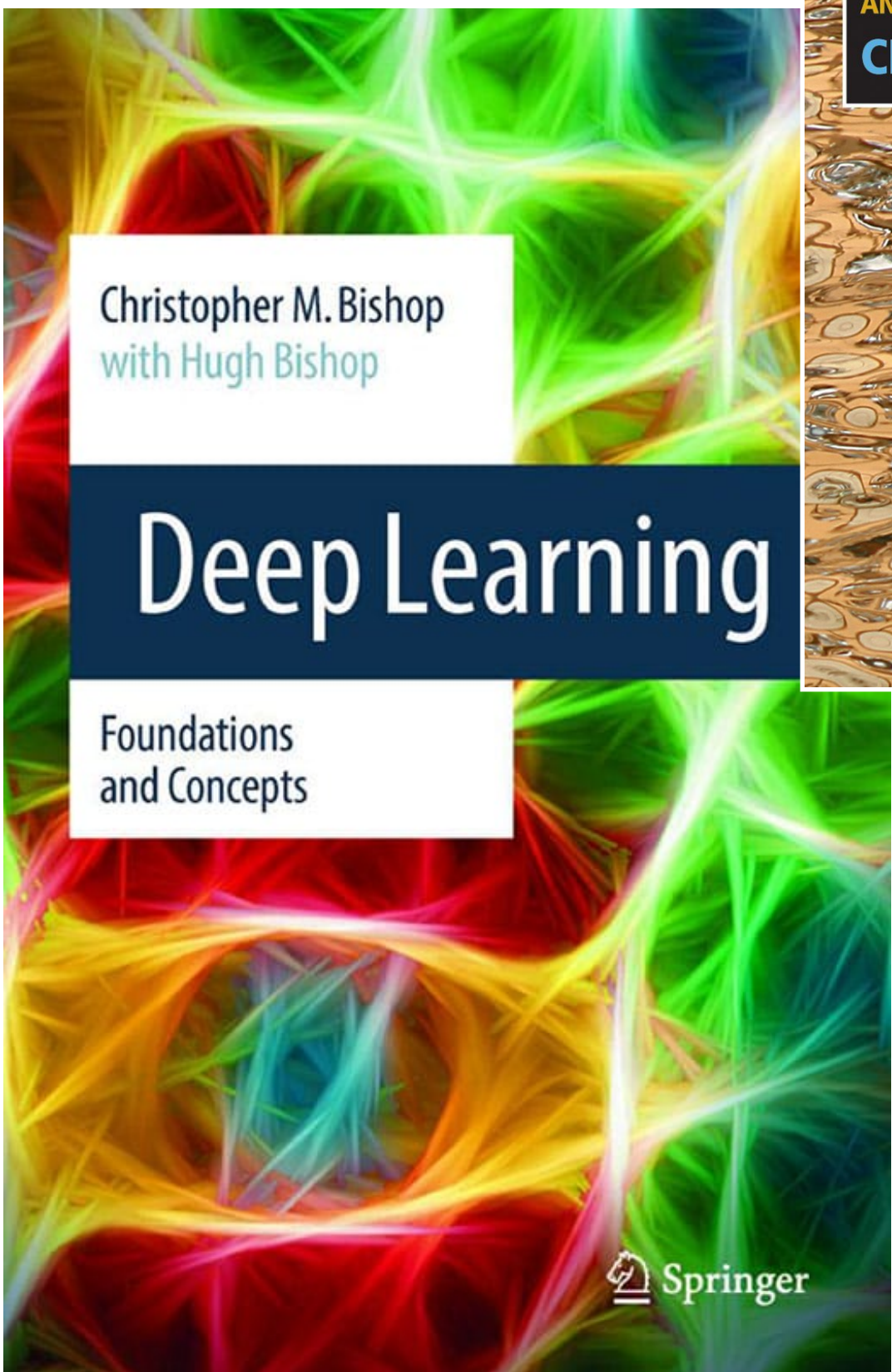
C = Chapter
S = Section
P = Page



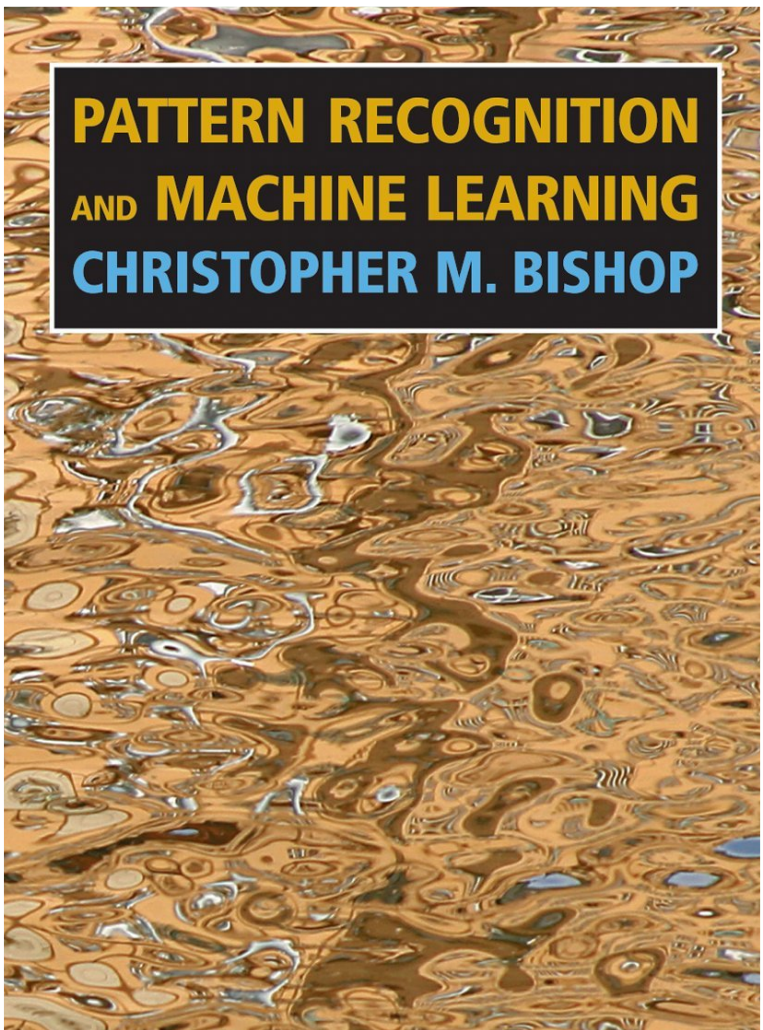
ISL
(OA)



ESL
(OA)



DL
(OA)



PRML

Others /

Shalev-Shwartz+(2014).
*Understanding machine learning:
From theory to algorithms*

Burkov(2019).
*The Hundred-page Machine
Learning Book*

Who are you?

Survey : Background & Skills

Feedback

Paramètres

Questions

Modèles

Réponses

Plus ▾

This short survey (10-15 min) helps us understand where you're starting from, so we can pitch explanations and examples at the right level for the group. It covers your background, math/ML/coding experience, and (if relevant) neuroscience exposure.

This survey is anonymous and has no impact on your grade. There are no right or wrong answers — "I've never done this" is just as useful to us as "I'm very comfortable with this."

Teaser

What's your guess for P(doom)?

When the WSJ starts asking existential questions, you know AI no longer fits in a statistics book

THE WALL STREET JOURNAL.

TECHNOLOGY • ARTIFICIAL INTELLIGENCE [Follow](#)

How Would AI Actually Kill Us All? What to Know About the AI Doomsday Debate

A growing chorus of AI researchers say the technology **could** wipe out humans—even if all the bots want to do is make paper clips

Sept. 9, 2026

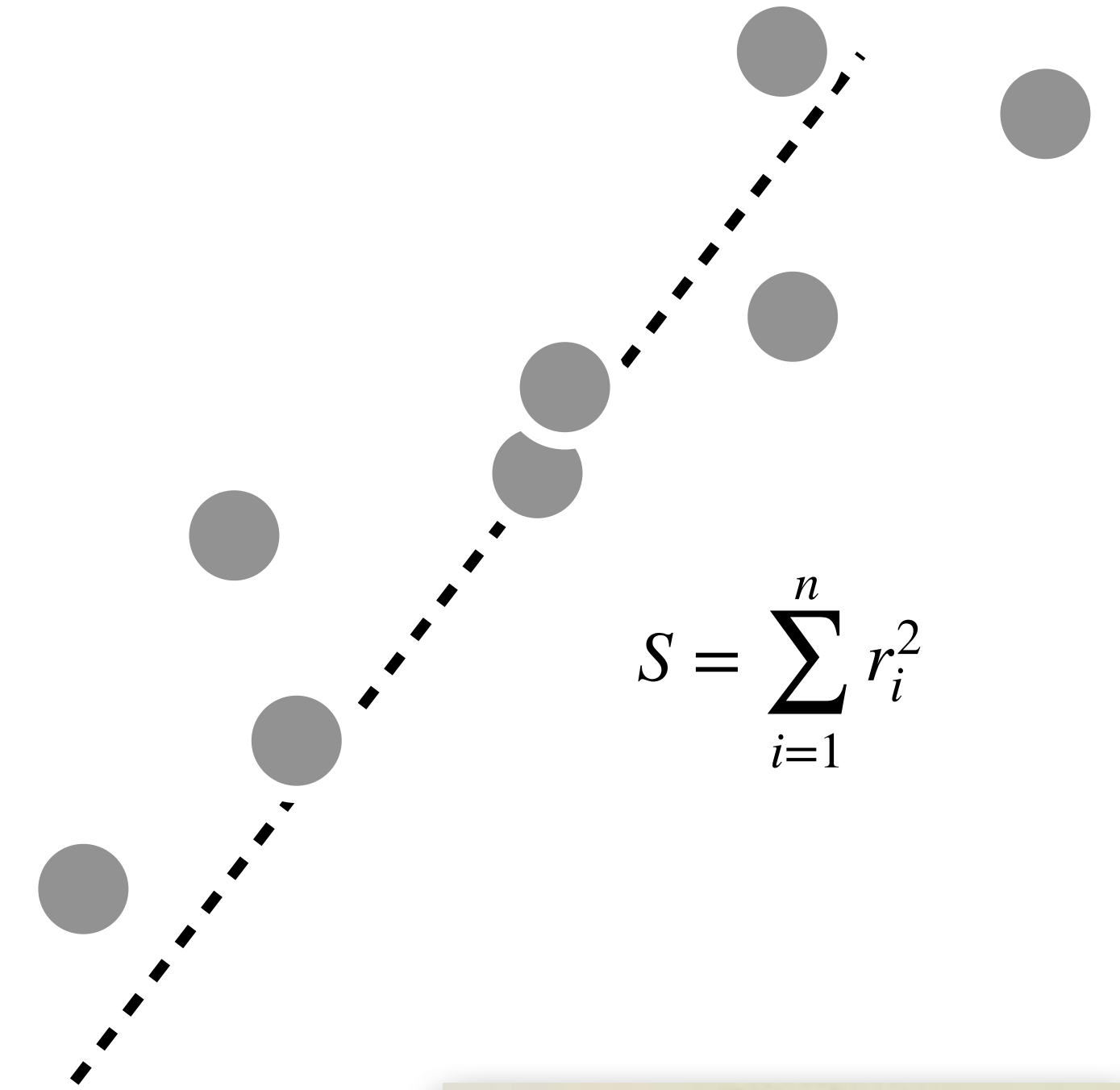
Uncertainty at this scale doesn't stay contained. It becomes public anxiety—for everyone

Even the builders don't fully understand the technology they built. Even less the one they will build

Machine Learning



Automatic discovery of regularities in data and their use to take action



A.-M. Legendre
(1752-1833)

Machine Learning

Anatomy of a Definition

The key is to **find algorithms**—that work, and that we can reason about

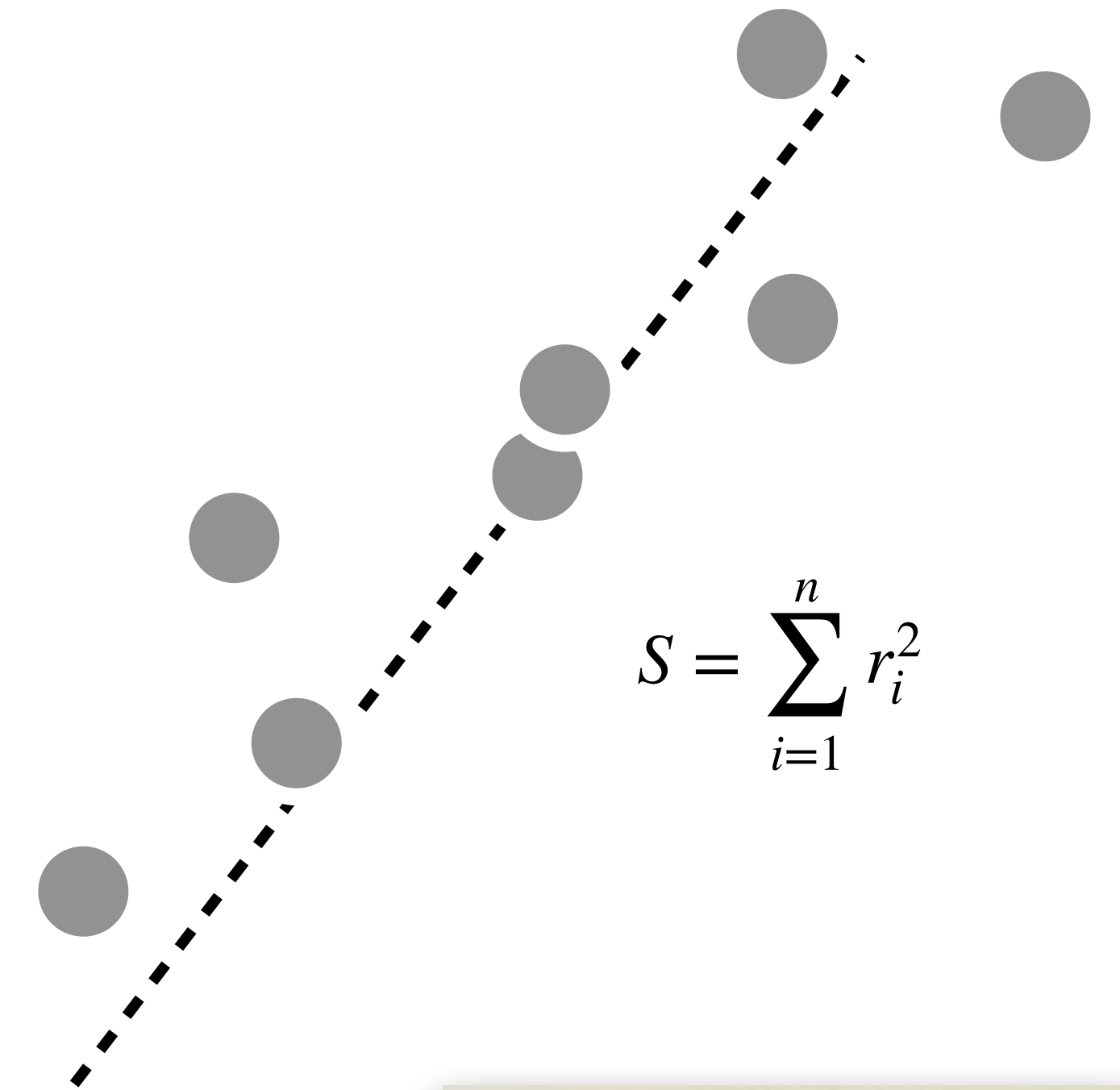
The world is not random, **there is structure** to « discover »



Automatic discovery of regularities in data and their use to take action

Data are **noisy measurements** of the world. The more, the better

This discipline is **goal-oriented**: every algorithm serves a purpose



$$S = \sum_{i=1}^n r_i^2$$



A.-M. Legendre
(1752-1833)

AI, ML & DL

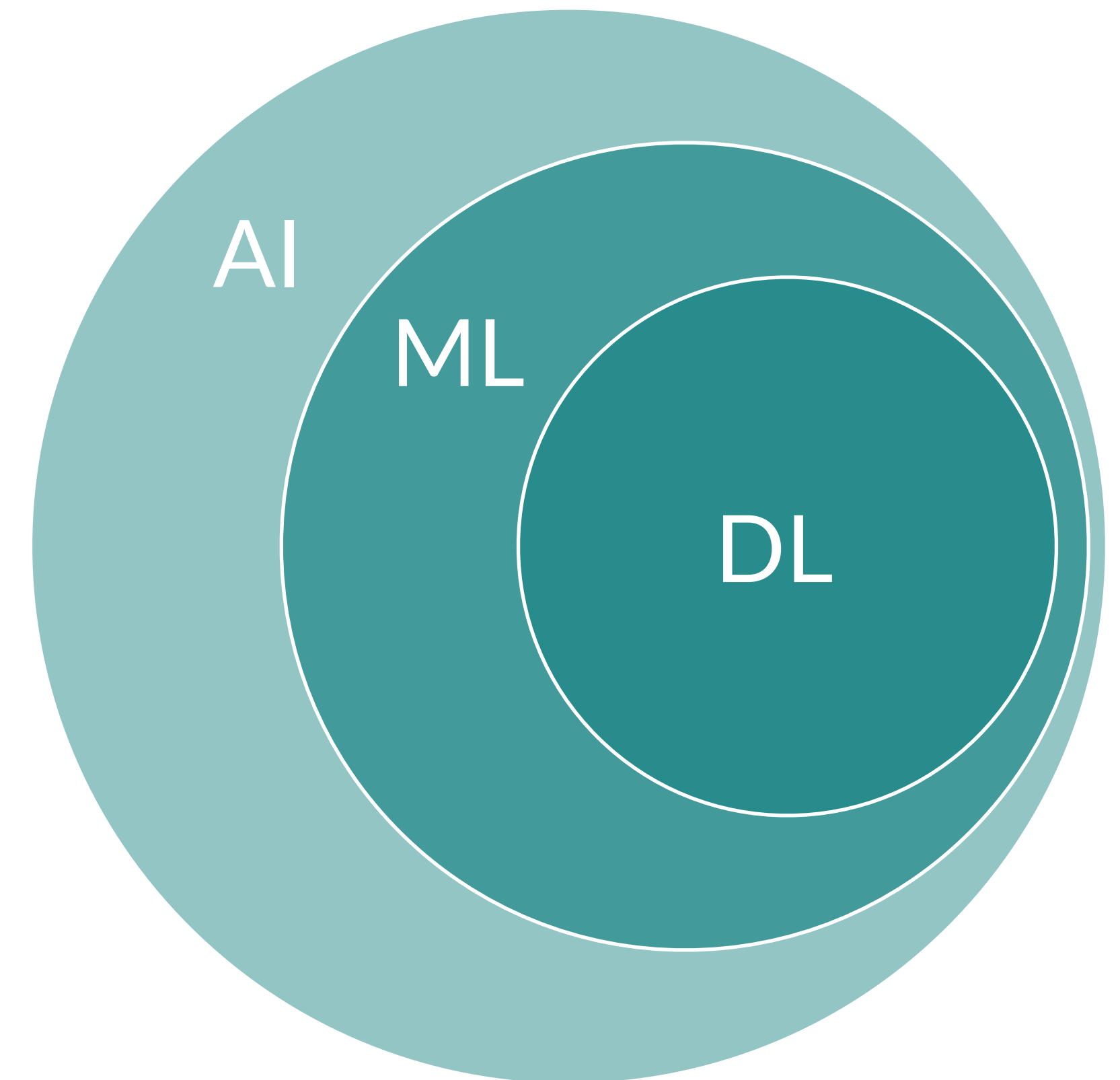
Artificial Intelligence / Creating systems that exhibit « intelligent » behaviour

Machine Learning / Learning from data through statistical pattern recognition

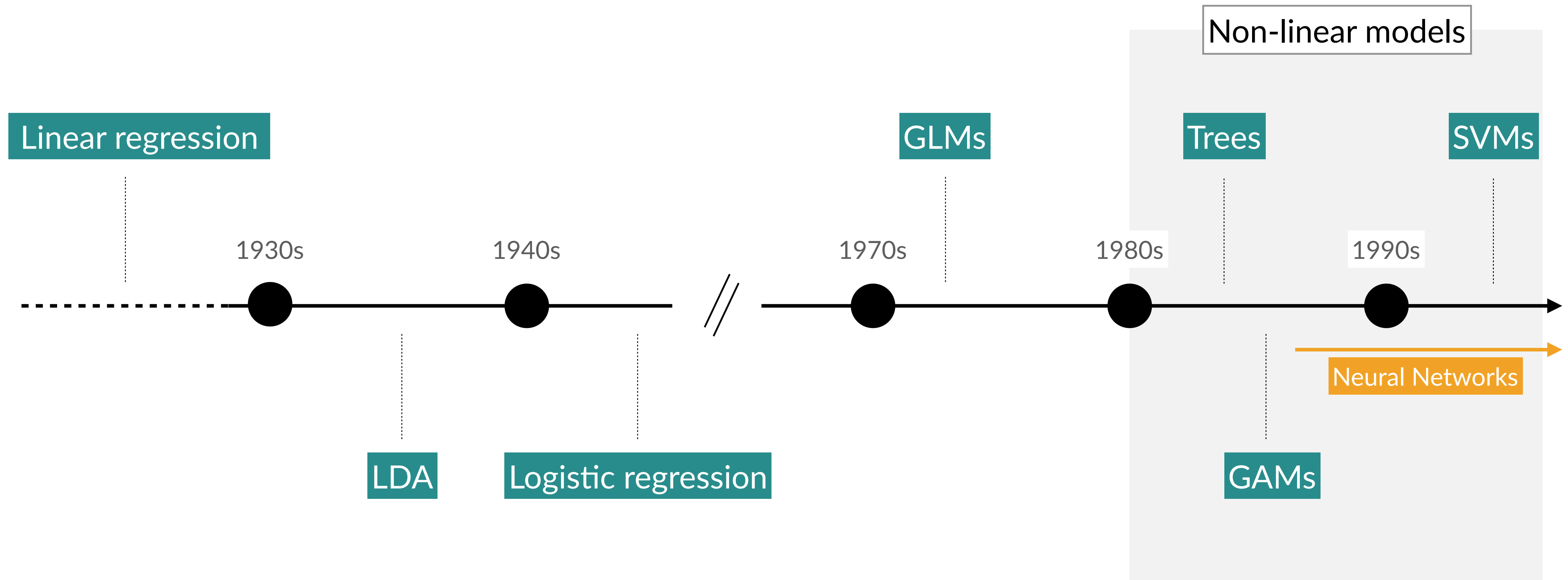
k-Nearest Neighbours (k-NN), Support Vector Machines (SVM) ...

Deep Learning / Multi-layered neural networks that automatically discover features

Multi-Layer Perceptron (MLP), Convolutional Neural Networks (CNN) ...

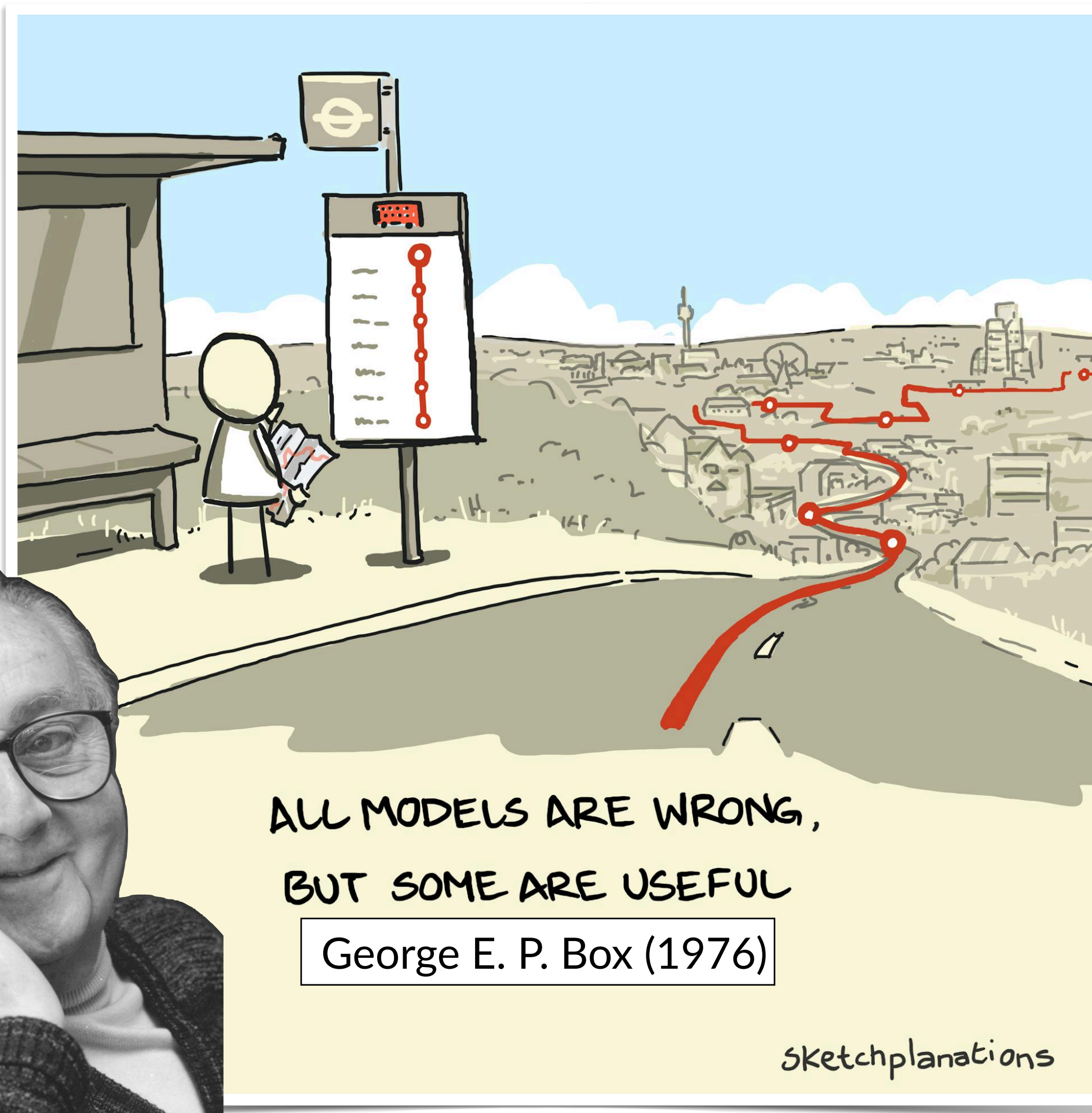


Brief History of Machine Learning



Some General Facts

The most quoted quote in ML



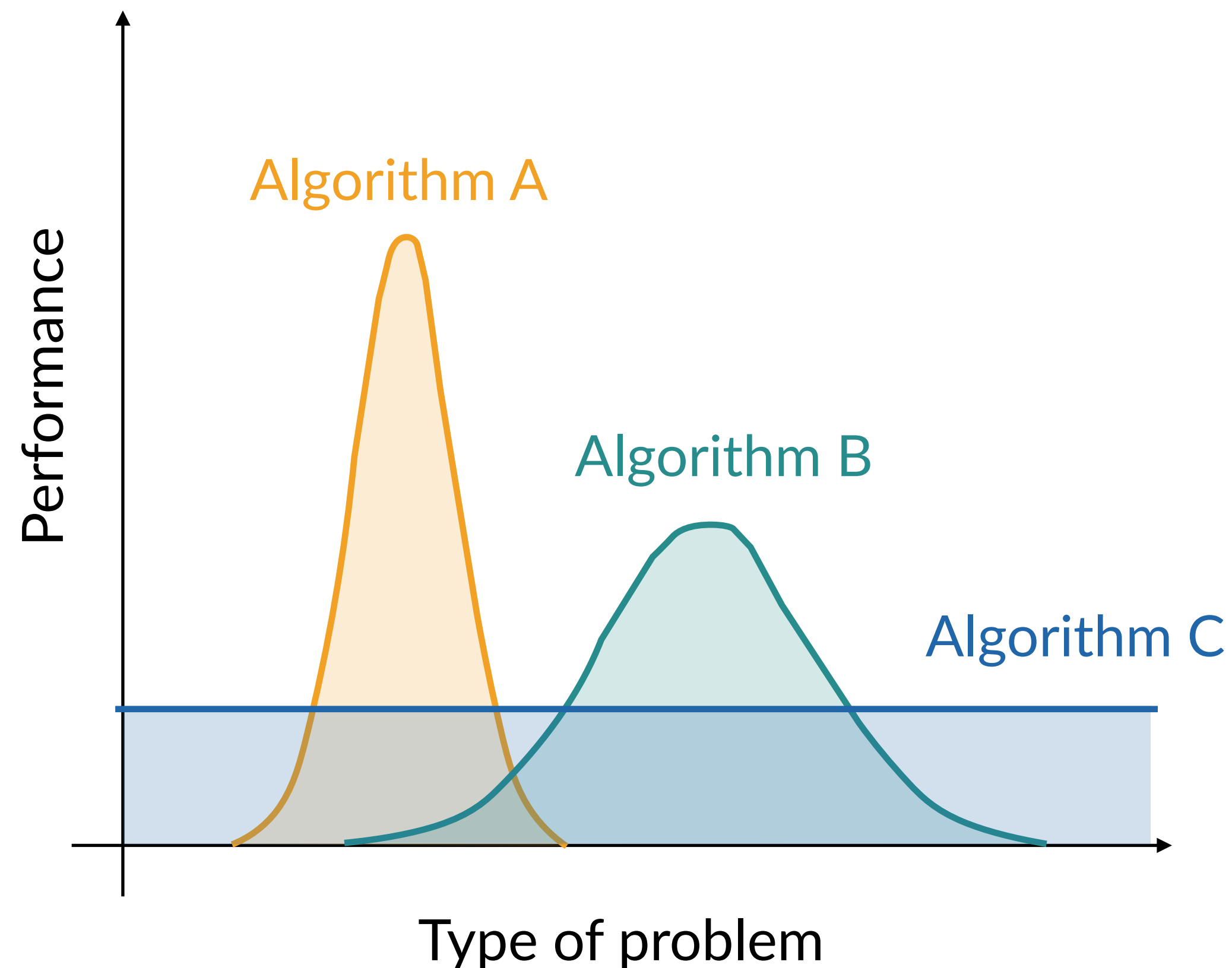
There is **no such a thing as the « true » model** of the world-out-there

The gap between our model and the world—noise, error, residual— isn't something to dismiss. It's something to measure, understand, and work with.

Learning to handle ignorance, precisely, is the key skill of a scientist

Some General Facts

No Free Lunch



No Free Lunch theorem (NFL) /

Every learning algorithm is as good as any other when averaged over all possible problems

Wolpert+(1996)

No universal recipe. Every method makes assumptions about the world and its performance depends on whether that assumption fits the problem

But for any given problem, some algorithms are clearly better suited than others!

Some General Facts

Predict or understand?

Given data, are you after an **accurate prediction**—or an **honest explanation**?

Often, you can't fully have both

Data Modelling Culture

Assume a stochastic model of how the data arose, fit it, and interpret its parameters

Statistical Learning
Pattern Recognition
« Classical » Machine Learning
(...)

Algorithmic Culture

Treat the learning mechanism as unknown, and just optimises for accurate prediction

Deep Learning

* This is a broad (and reductive) classification—plenty of research areas sit in between. If you work in one of these fields, please don't email me angrily

Statistical Science
2001, Vol. 16, No. 3, 199–231

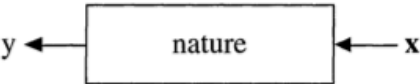
Statistical Modeling: The Two Cultures

Leo Breiman

Abstract. There are two cultures in the use of statistical modeling to reach conclusions from data. One assumes that the data are generated by a given stochastic data model. The other uses algorithmic models and treats the data mechanism as unknown. The statistical community has been committed to the almost exclusive use of data models. This commitment has led to irrelevant theory, questionable conclusions, and has kept statisticians from working on a large range of interesting current problems. Algorithmic modeling, both in theory and practice, has developed rapidly in fields outside statistics. It can be used both on large complex data sets and as a more accurate and informative alternative to data modeling on smaller data sets. If our goal as a field is to use data to solve problems, then we need to move away from exclusive dependence on data models and adopt a more diverse set of tools.

1. INTRODUCTION

Statistics starts with data. Think of the data as being generated by a black box in which a vector of input variables $\mathbf{x}$ (independent variables) go in one side, and on the other side the response variables $\mathbf{y}$ come out. Inside the black box, nature functions to associate the predictor variables with the response variables, so the picture is like this:



There are two goals in analyzing the data:

Prediction. To be able to predict what the responses are going to be to future input variables;
Information. To extract some information about how nature is associating the response variables to the input variables.

There are two different approaches toward these goals:

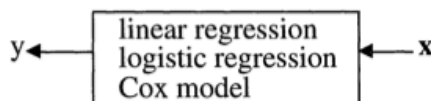
The Data Modeling Culture

The analysis in this culture starts with assuming a stochastic data model for the inside of the black box. For example, a common data model is that data are generated by independent draws from

response variables = $f(\text{predictor variables, random noise, parameters})$

Leo Breiman is Professor, Department of Statistics, University of California, Berkeley, California 94720-4735 (e-mail: leo@stat.berkeley.edu).

The values of the parameters are estimated from the data and the model then used for information and/or prediction. Thus the black box is filled in like this:

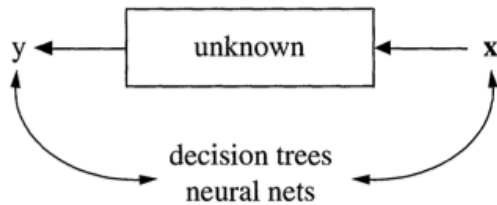


Model validation. Yes—no using goodness-of-fit tests and residual examination.

Estimated culture population. 98% of all statisticians.

The Algorithmic Modeling Culture

The analysis in this culture considers the inside of the box complex and unknown. Their approach is to find a function $f(\mathbf{x})$ —an algorithm that operates on $\mathbf{x}$ to predict the responses $\mathbf{y}$. Their black box looks like this:



Model validation. Measured by predictive accuracy.
Estimated culture population. 2% of statisticians, many in other fields.

In this paper I will argue that the focus in the statistical community on data models has:

- Led to irrelevant theory and questionable scientific conclusions;

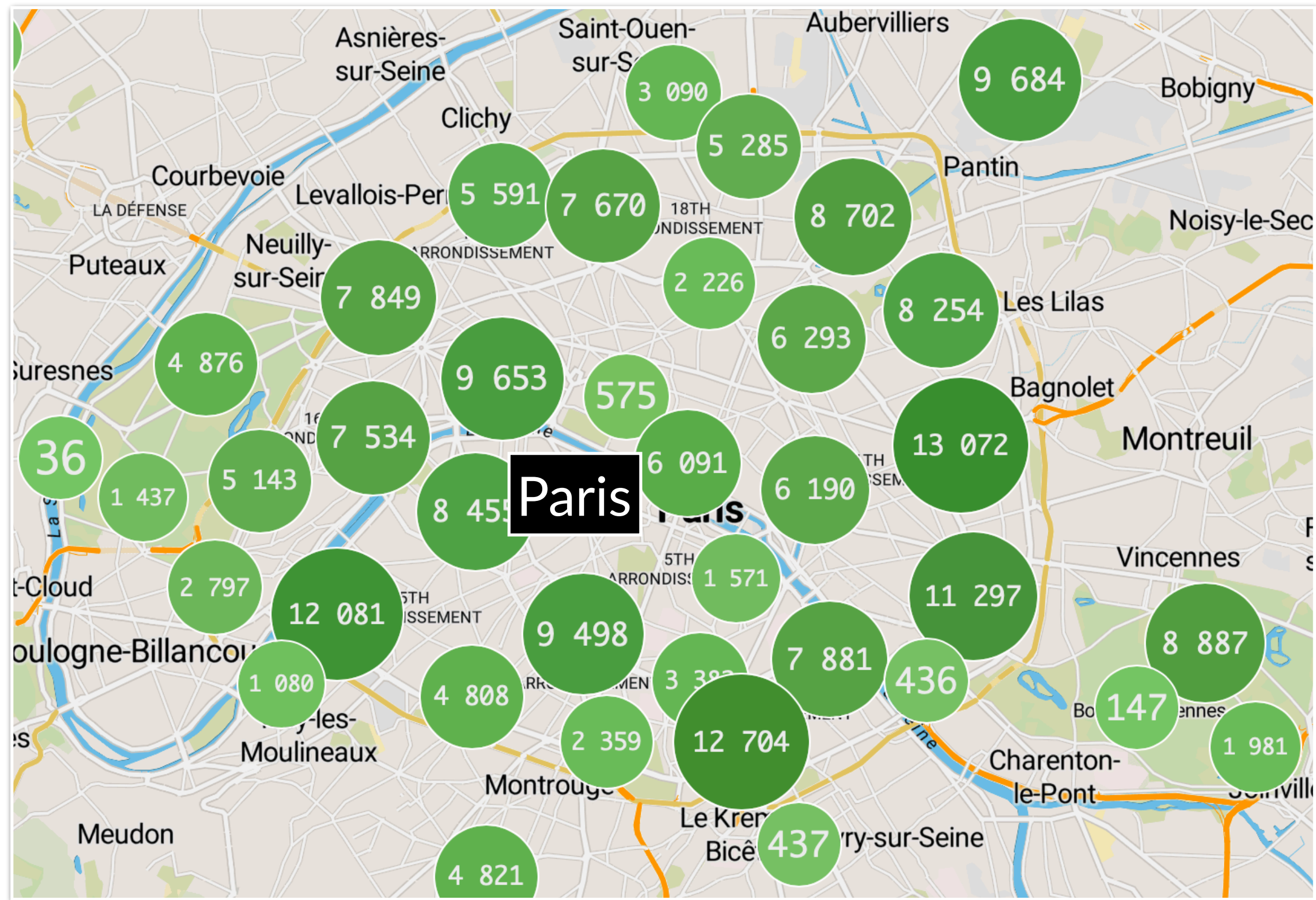
L Breiman (2001)

A Refresher on Probability

The Trees of Paris

les-arbres /

Paris's public tree inventory—every tree the city manages, geolocated, with species, size, and location type recorded



$n = 220\,022$ trees

From [Paris OpenData](#),
Last update: 11 septembre 2026



The Trees of Paris

Dataset **les-arbres** : Paris's public tree inventory—every tree the city manages, geolocated, with species, size, and location type recorded

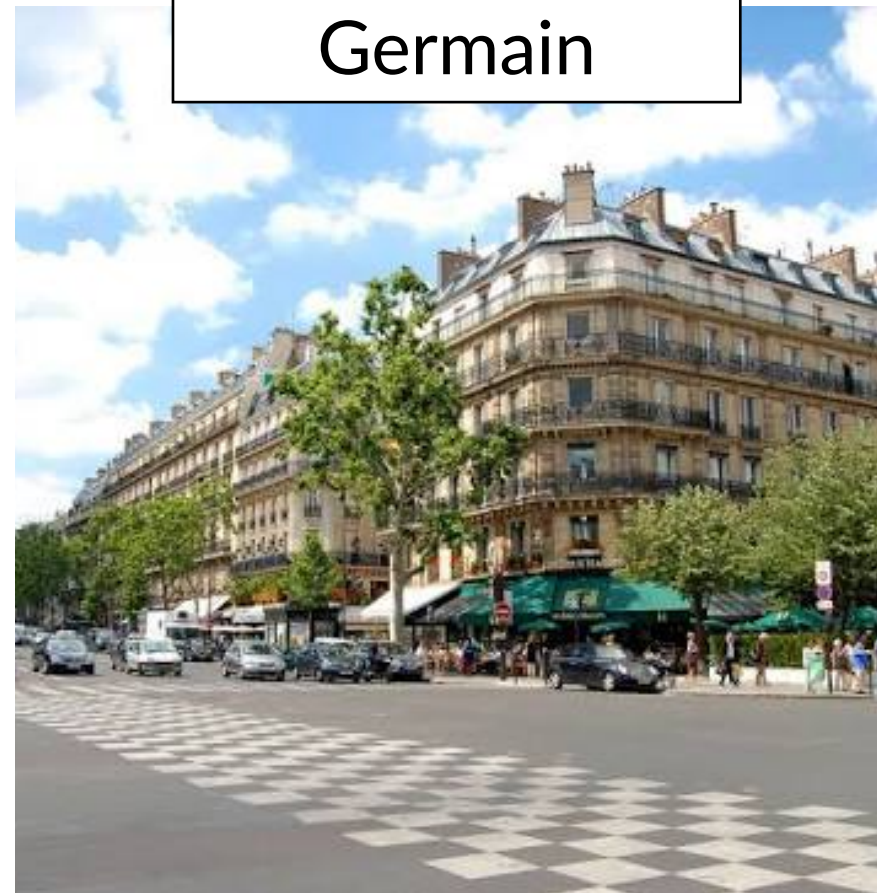
From [Paris OpenData](#)

$\tilde{X} \in \mathbb{R}$: Circumference (cm)
(Continuous)



$Y \in \{\text{street, park}\}$: Location
(Categorical)

Boulevard Saint-Germain



Parc des Buttes Chaumont



What is Probability?

A Frequentist View

Q / If you picked a random tree in Paris, what is the probability that it's located on a street?

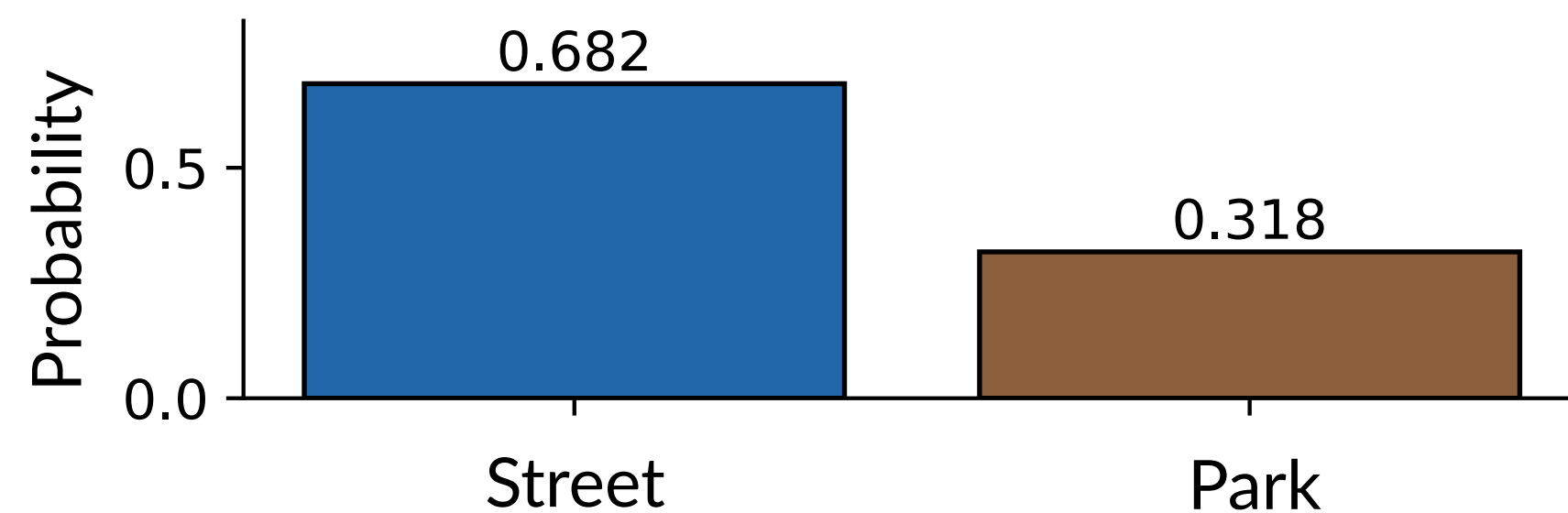
Random variable

Number of occurrences of y in the dataset

$$P(Y = y) \simeq \frac{n_y}{n}$$

« street » or « park »

Size of the dataset



What is Probability?

A Frequentist View

Q / If you picked a random tree in Paris, what is the probability that it's located on a street?

Q / If you picked a random tree in Paris, what is the probability that its circumference is 80.90862 cm?

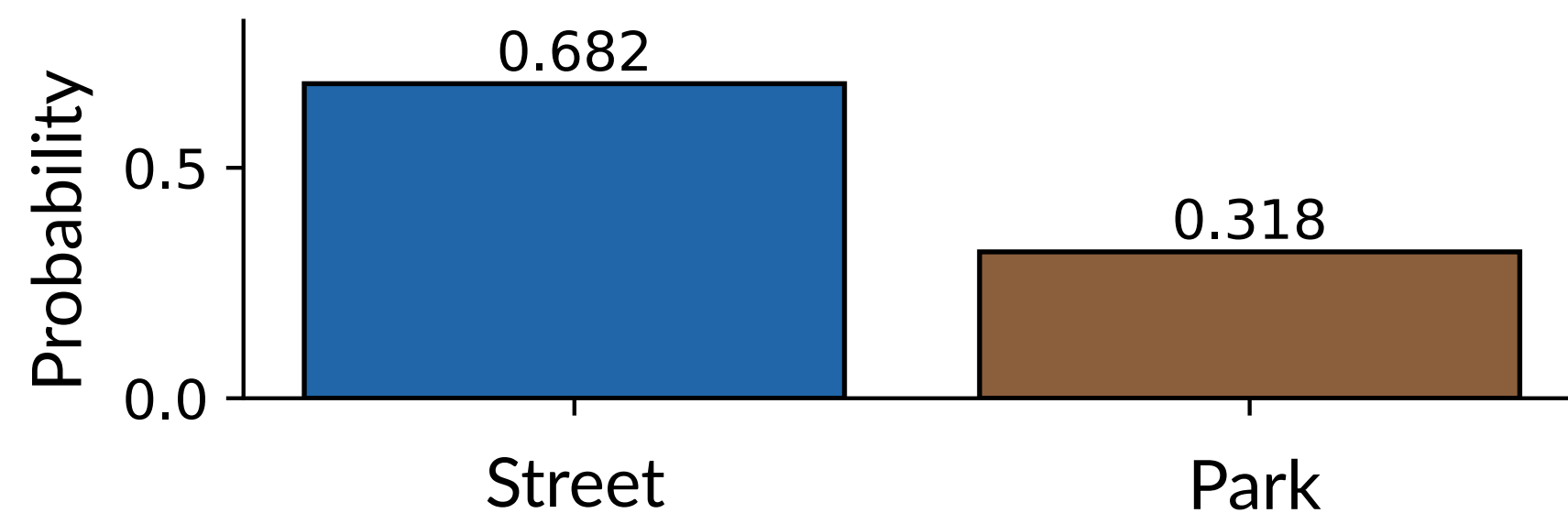
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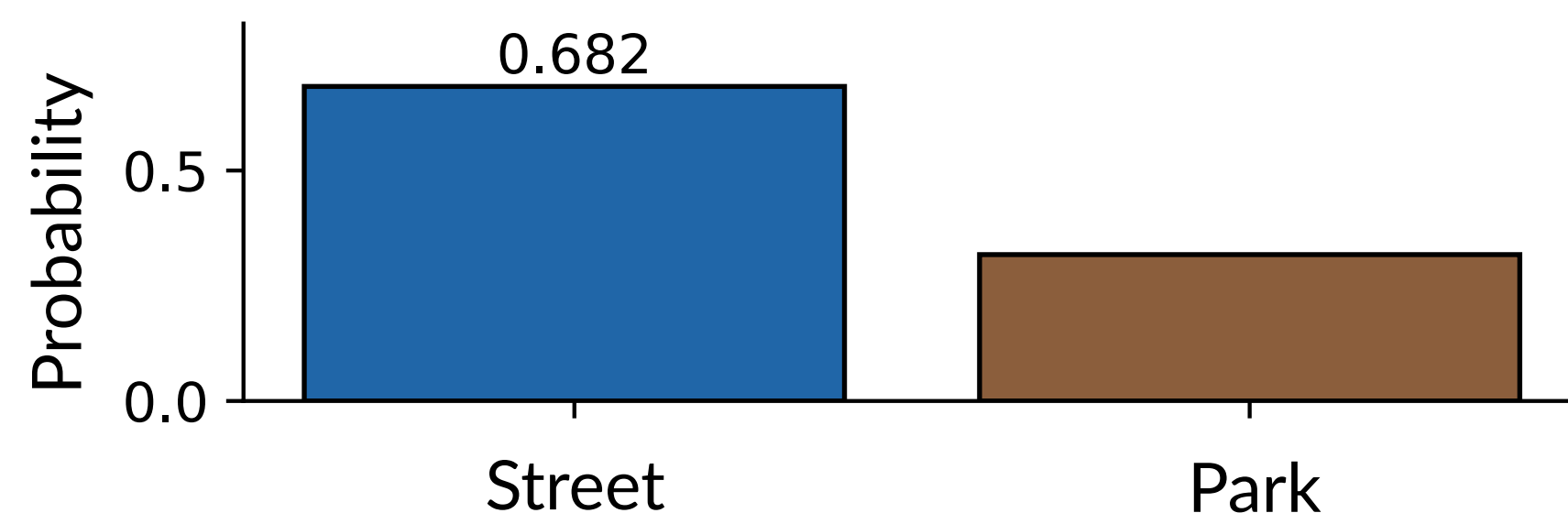
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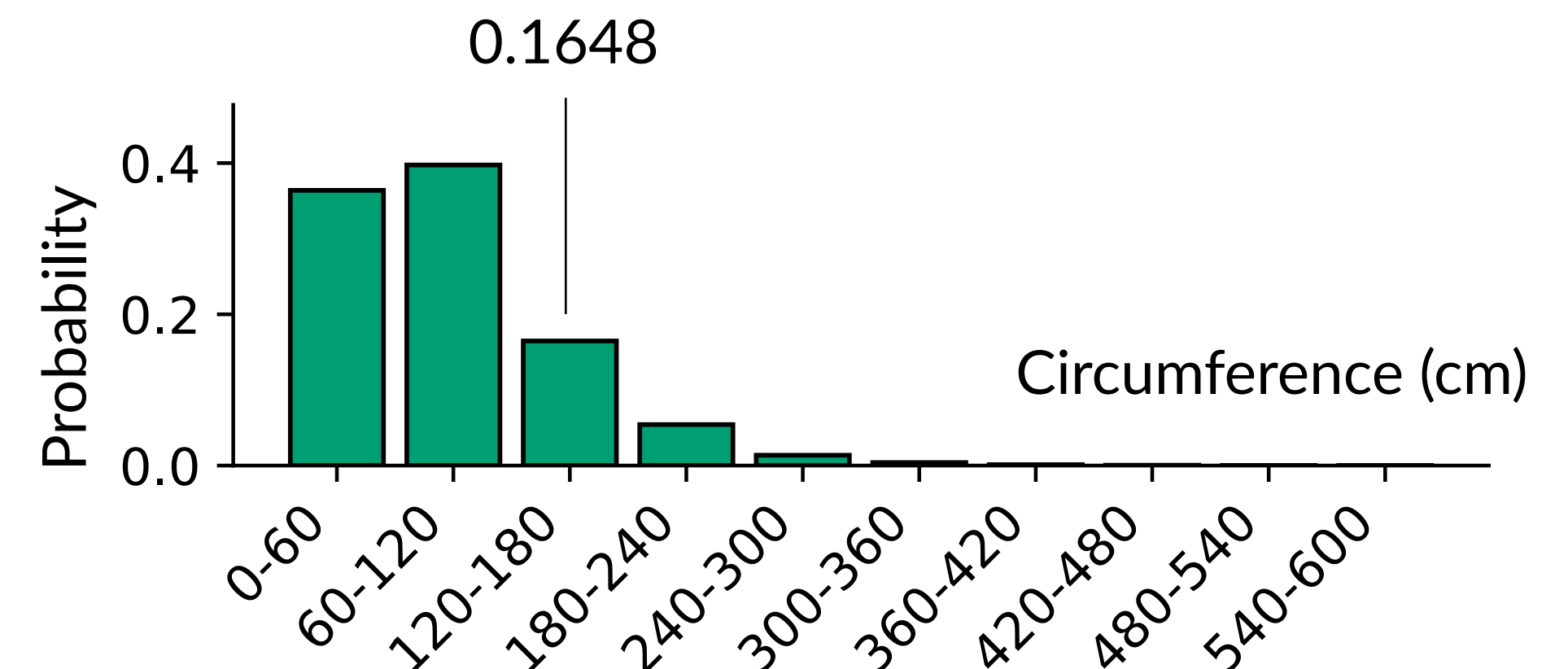
Size of the dataset



Q / If you picked a random tree in Paris, what is the probability that its circumference is ~~80.90862 cm~~?

between 120 and 180 cm?

$$P(X = x) \simeq \frac{n_x}{n} \quad \left| \begin{array}{l} X \text{ is the discretised (or} \\ \text{binned)} \text{ version of } \tilde{X} \end{array} \right.$$

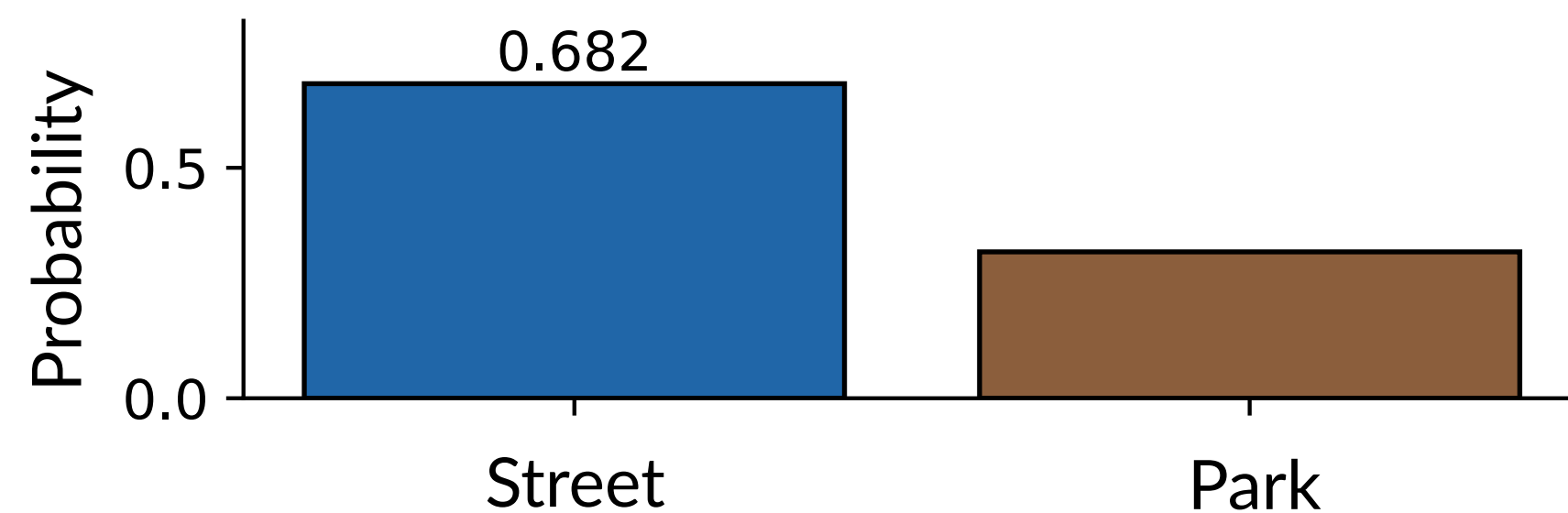


What is Probability?

A Frequentist View

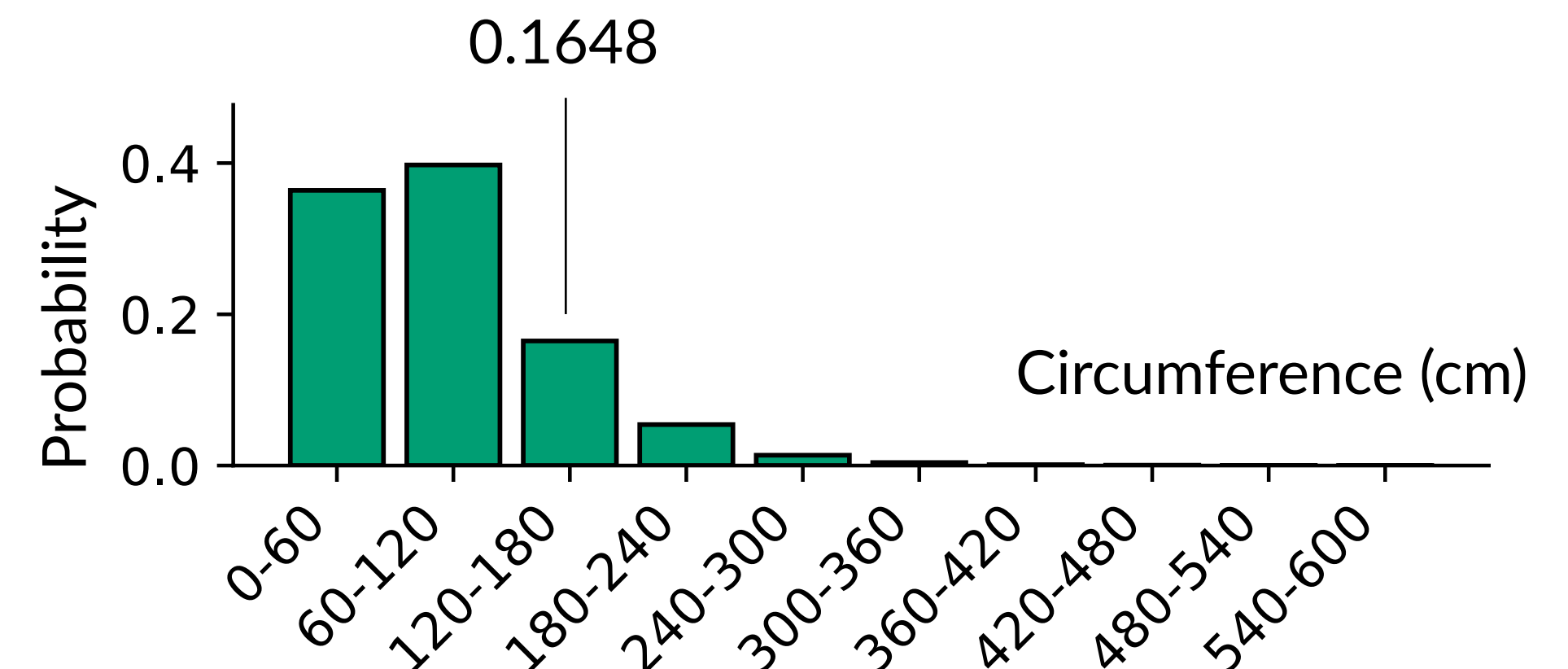
Q / If you picked a random tree in Paris, what is the probability that it's located on a street?

$$\sum_j P(Y = y_j) = 1$$



Q / If you picked a random tree in Paris, what is the probability that its circumference is ~~80.90862 cm~~ between 120 and 180 cm?

$$\sum_i P(X = x_i) = 1$$



The Grammar of Probability

Joint Probability

Q / Does where a tree grows tell you anything about how big it is?

→ To answer that, we need to **look at X and Y together**—not one at a time

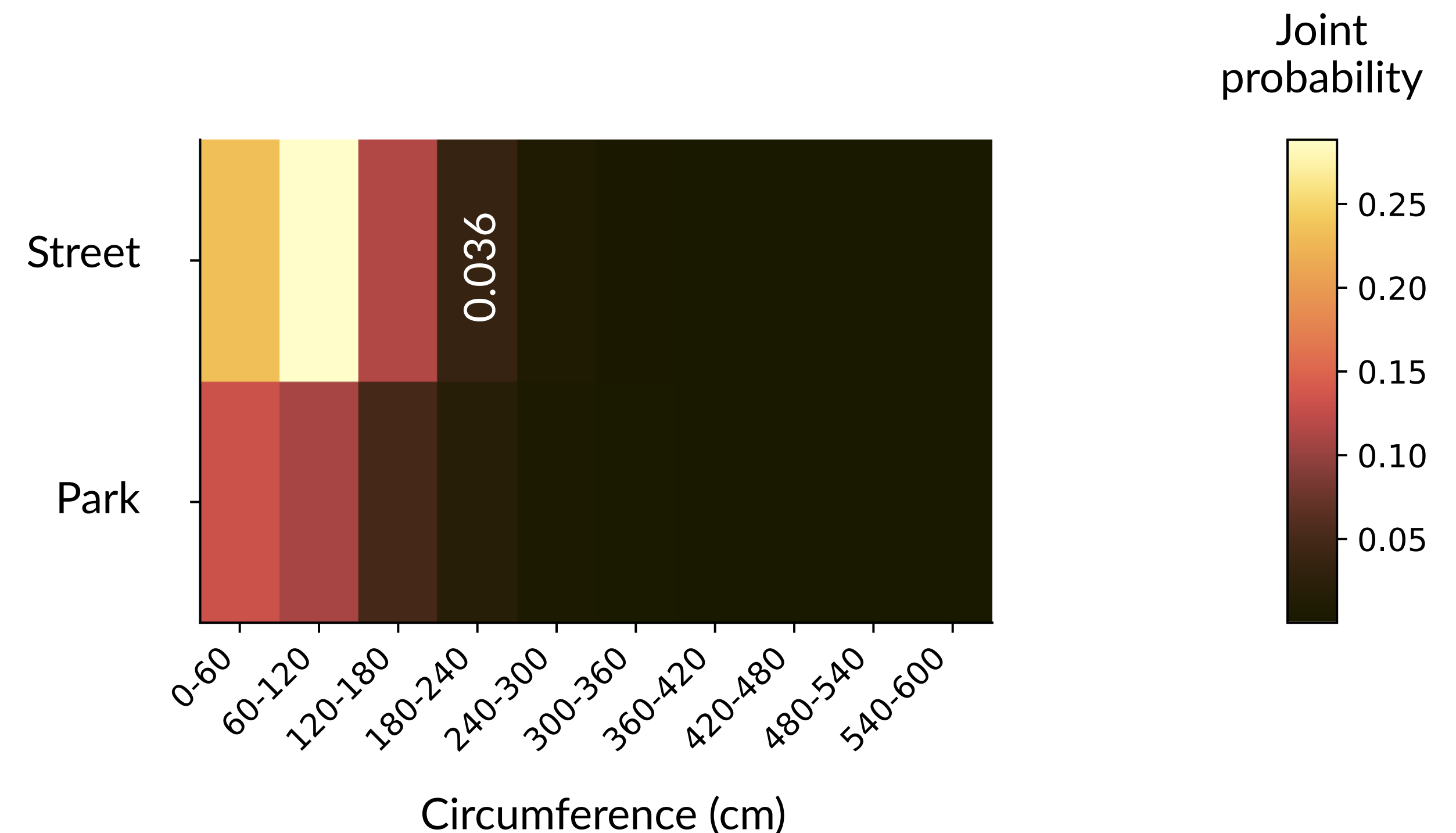
Joint Probability

$$P(X = x_i, Y = y_j) \simeq \frac{n_{ij}}{n}$$

« $P(X, Y)$ »

$$\sum_{ij} P(X = x_i, Y = y_j) = 1$$

For instance, the probability that a randomly selected tree has a circumference between 180 and 240cm and is located on a street is 3.6%



The Grammar of Probability

Sum Rule

Marginal probability: the probability of one variable, ignoring the other entirely

Sum rule

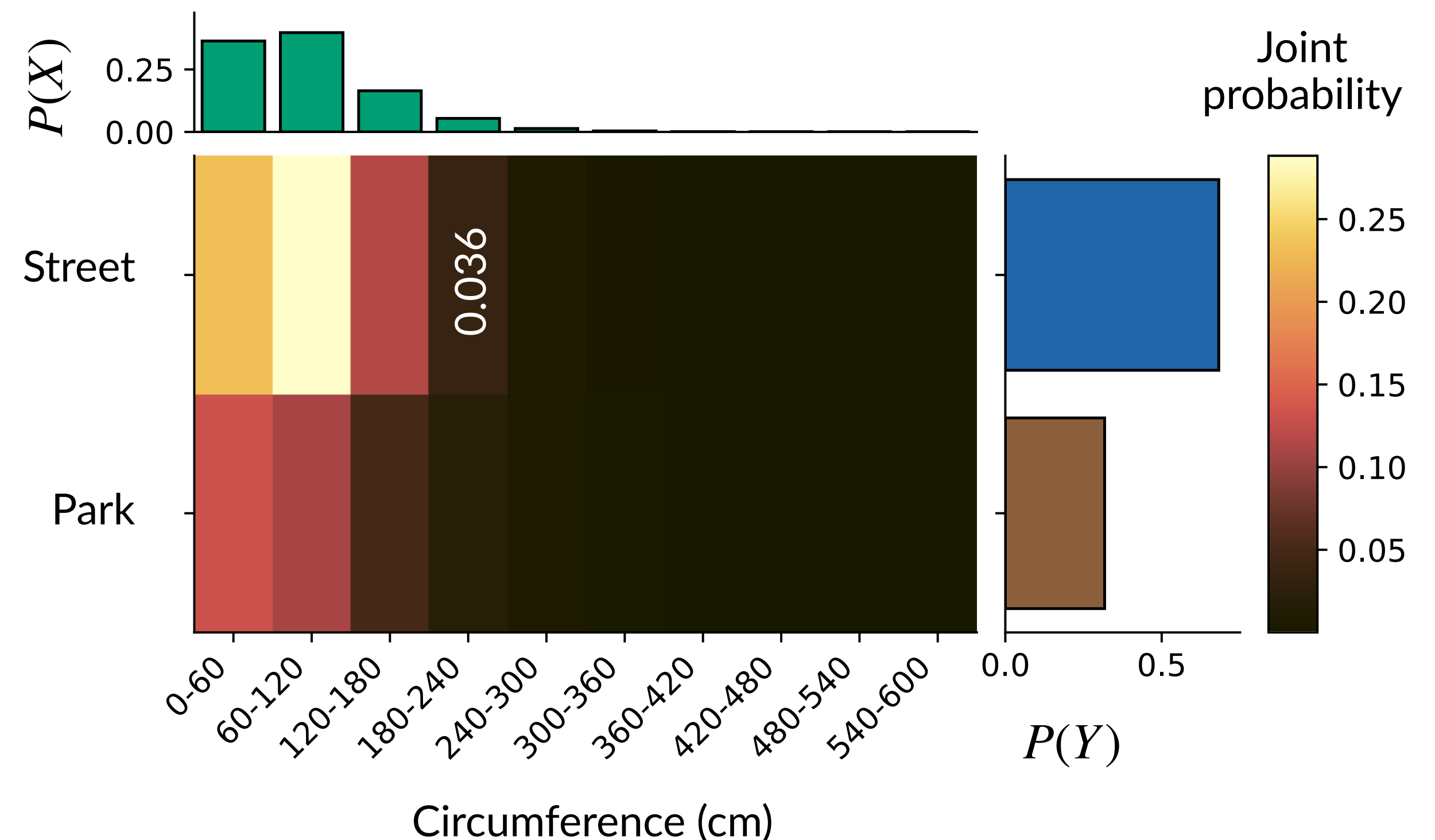
$$P(X) = \sum_Y P(X, Y)$$

$$P(Y) = \sum_X P(X, Y)$$

Summing out the variable we don't care about—hence « marginal.

* compact notation

For instance, the probability that a randomly selected tree has a circumference between 180 and 240cm and is located on a street is 3.6%



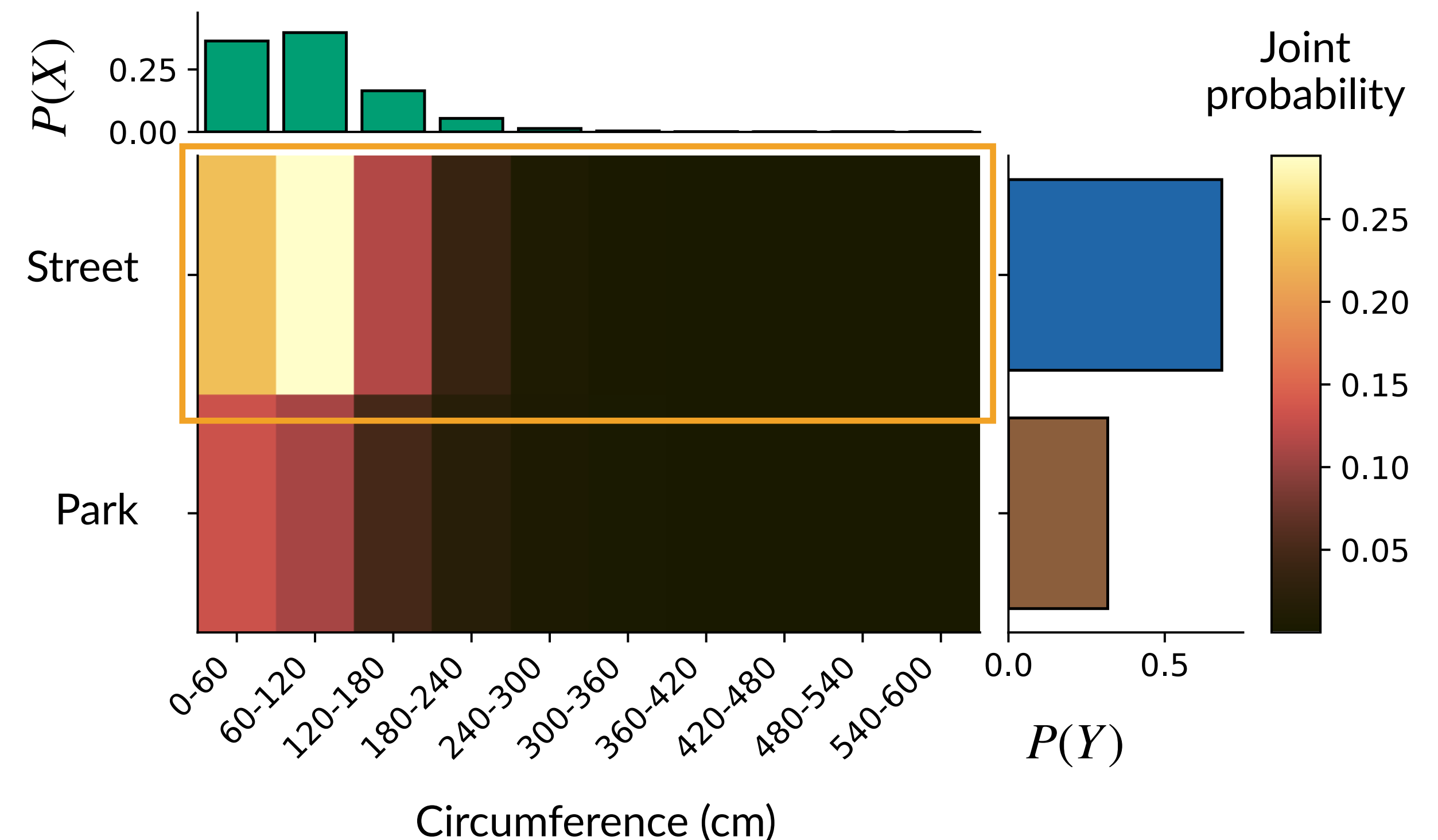
The Grammar of Probability

Conditional Probability

Q / What if we already know a tree is on a street—does that change what we expect about its size?

Conditional probability: the probability of one variable, given that we already know the other

$$\begin{aligned}
 P(X = x_i | Y = y_j) &\simeq \frac{n_{ij}}{\sum_i n_{ij}} \\
 \text{« } P(X | Y) \text{ »} & \\
 &= \frac{n_{ij}}{n} \frac{n}{\sum_i n_{ij}} \\
 &= P(X = x_i, Y = y_j) \frac{1}{P(Y = y_j)}
 \end{aligned}$$



The Grammar of Probability

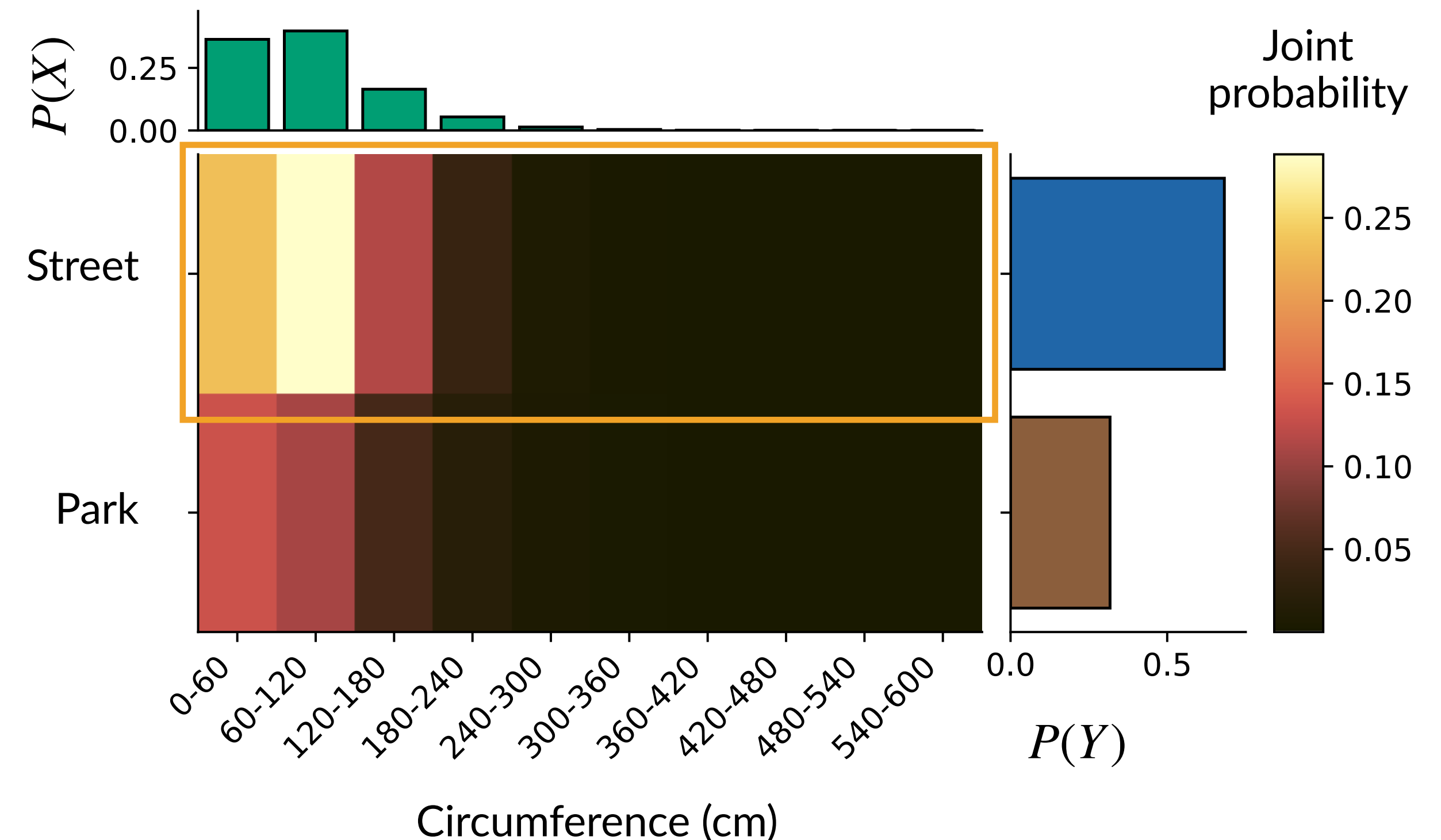
Product Rule

Q / What if we already know a tree is on a street—does that change what we expect about its size?

Conditional probability: the probability of one variable, given that we already know the other

Product rule

$$\begin{aligned} P(X, Y) &= P(X | Y)P(Y) \\ &= P(Y | X)P(X) \end{aligned}$$



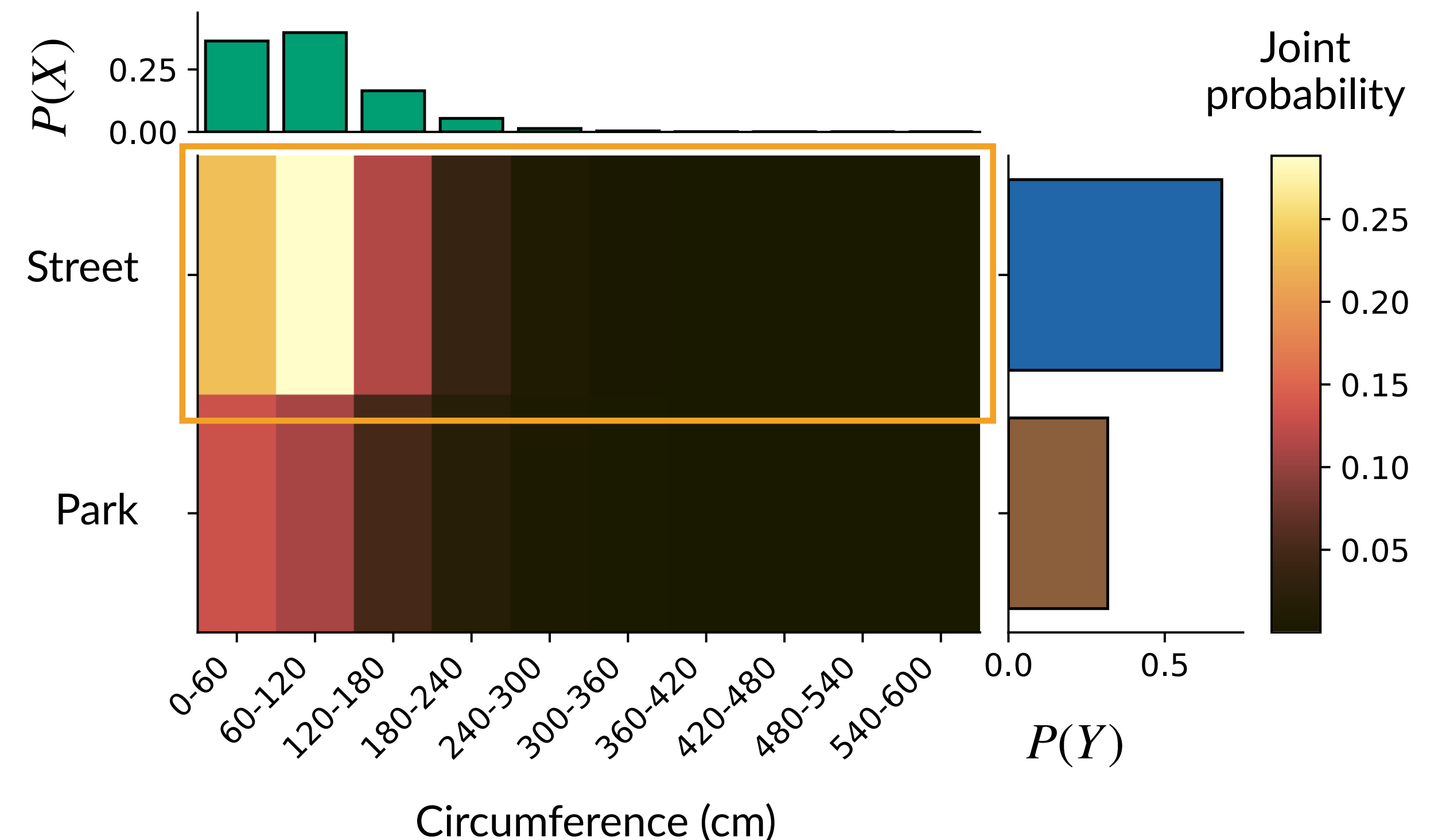
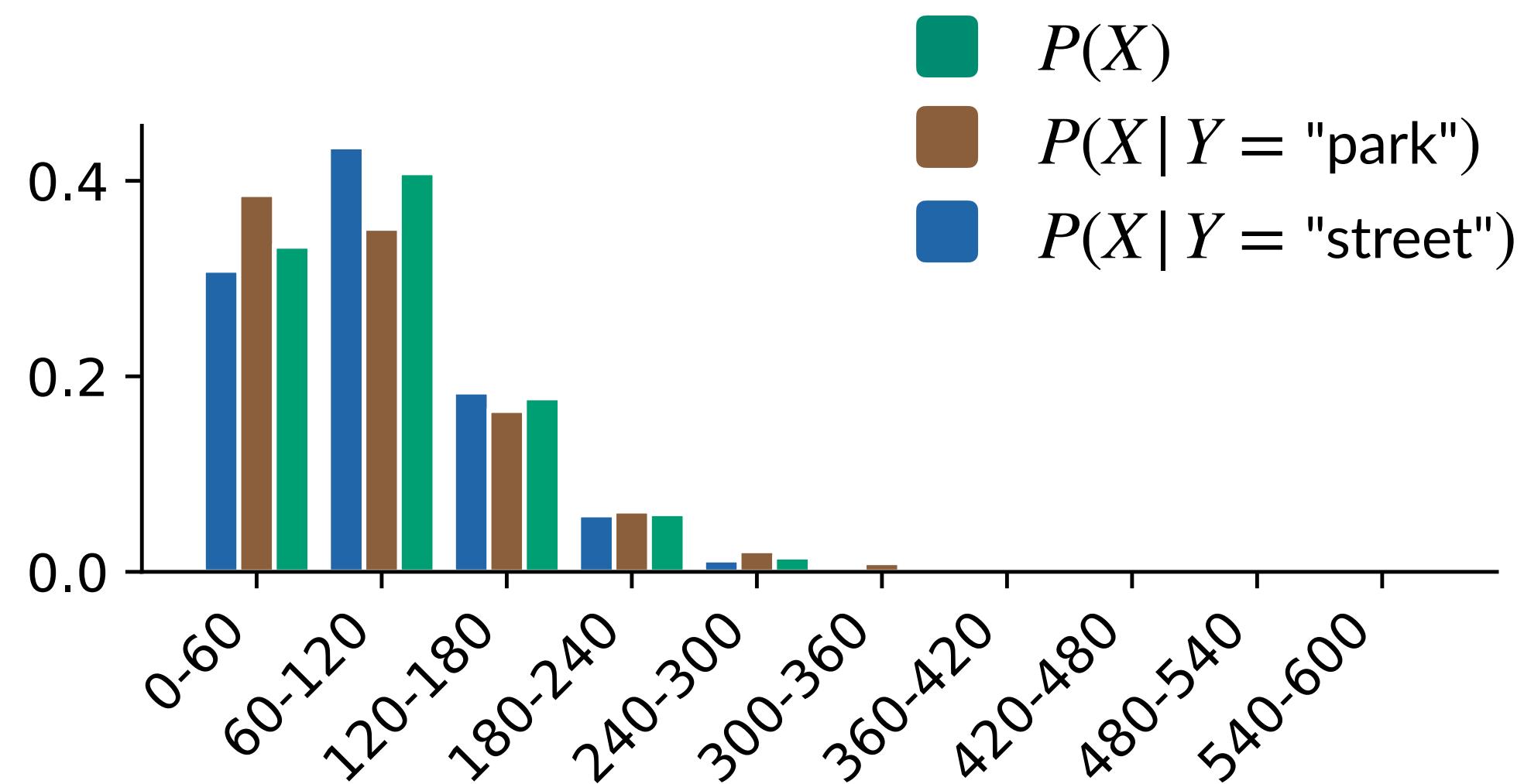
* compact notation

The Grammar of Probability

Product Rule

Q / What if we already know a tree is on a street—does that change what we expect about its size?

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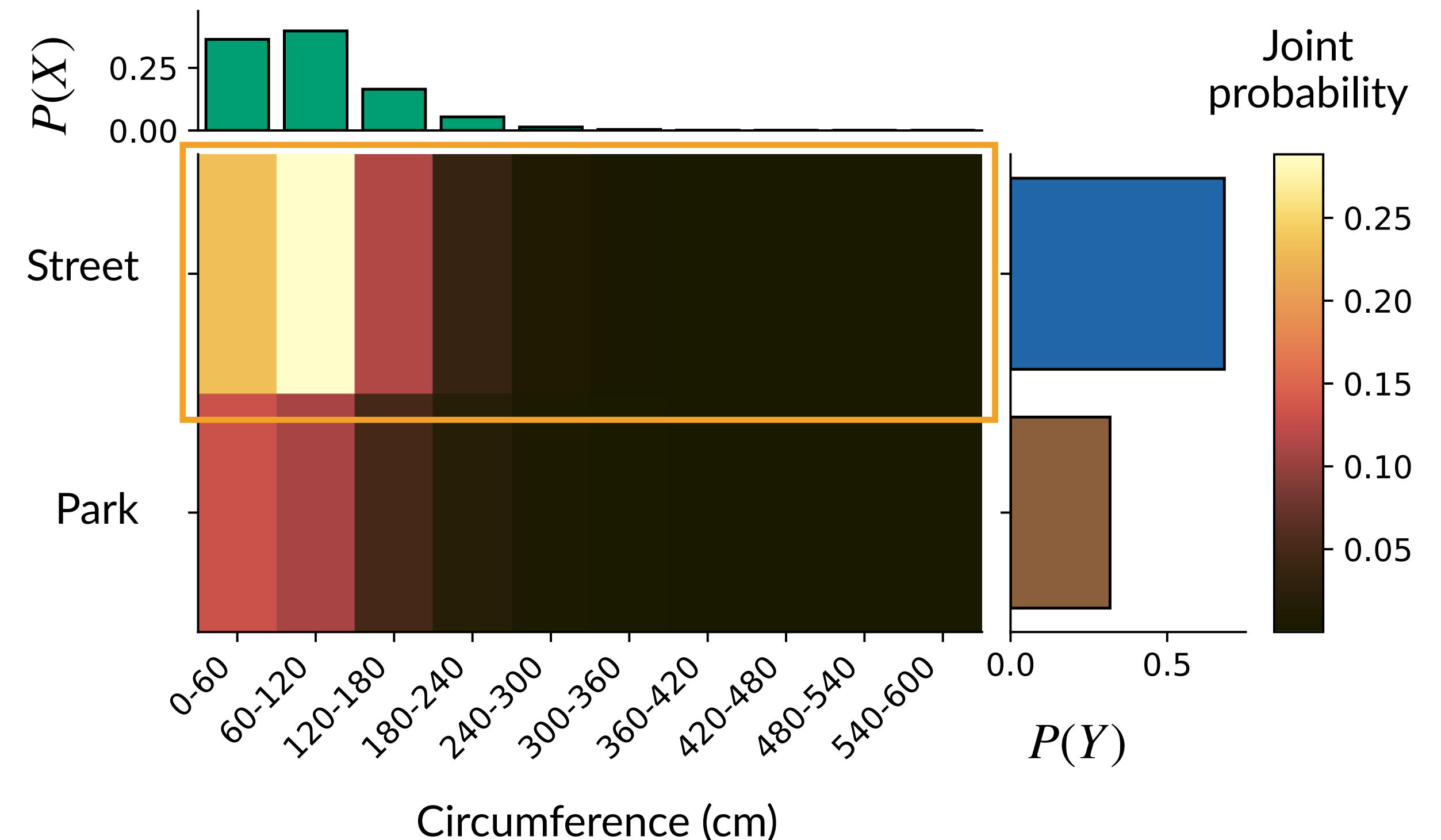
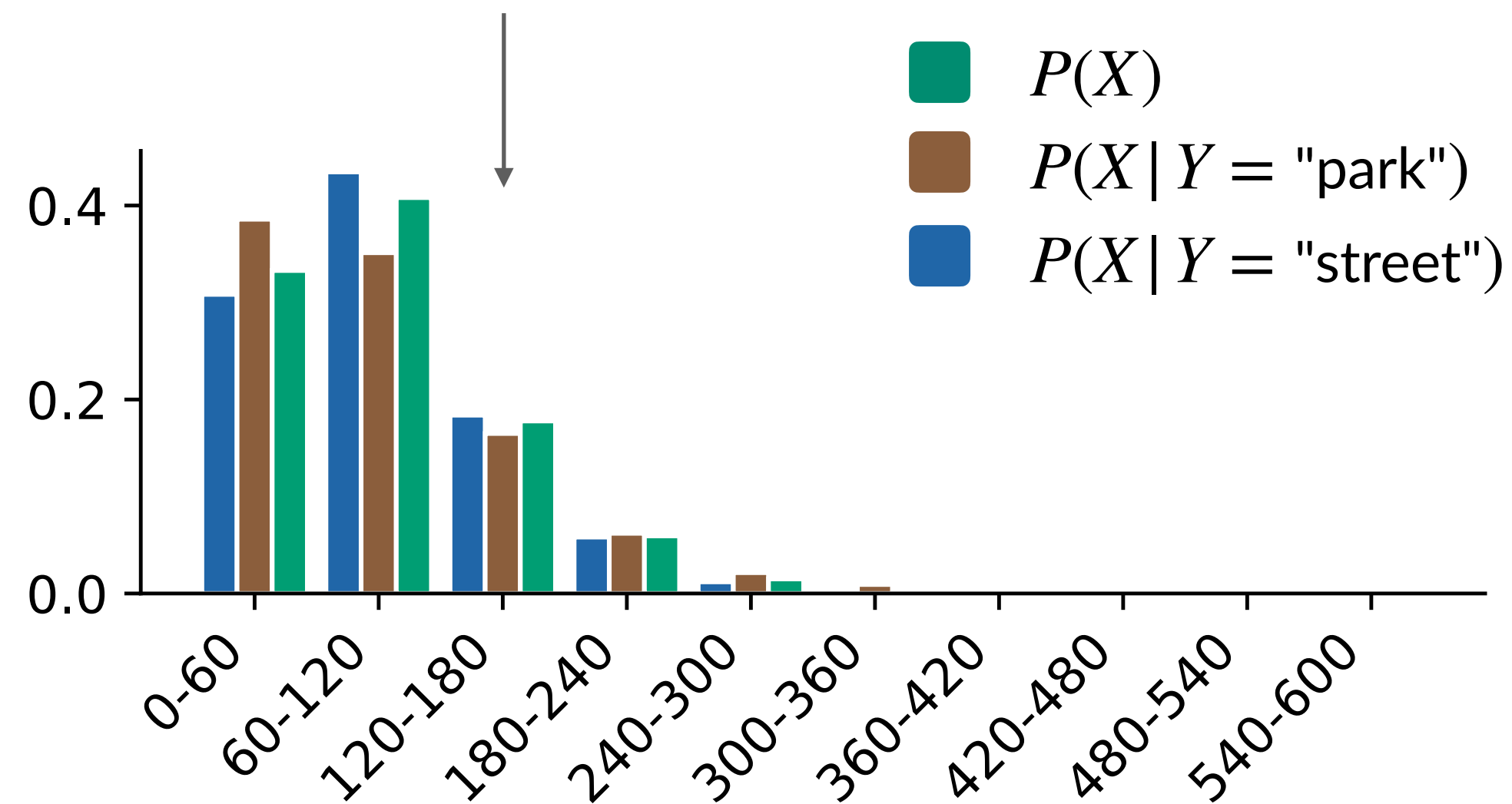


The Grammar of Probability

Independence

Two random variables are said to be **independent** if $P(X, Y) = P(X)P(Y)$ or, equivalently, if $P(X | Y) = P(X)$ and vice versa.

Here, X and Y are dependent—otherwise these three histograms would be the same!



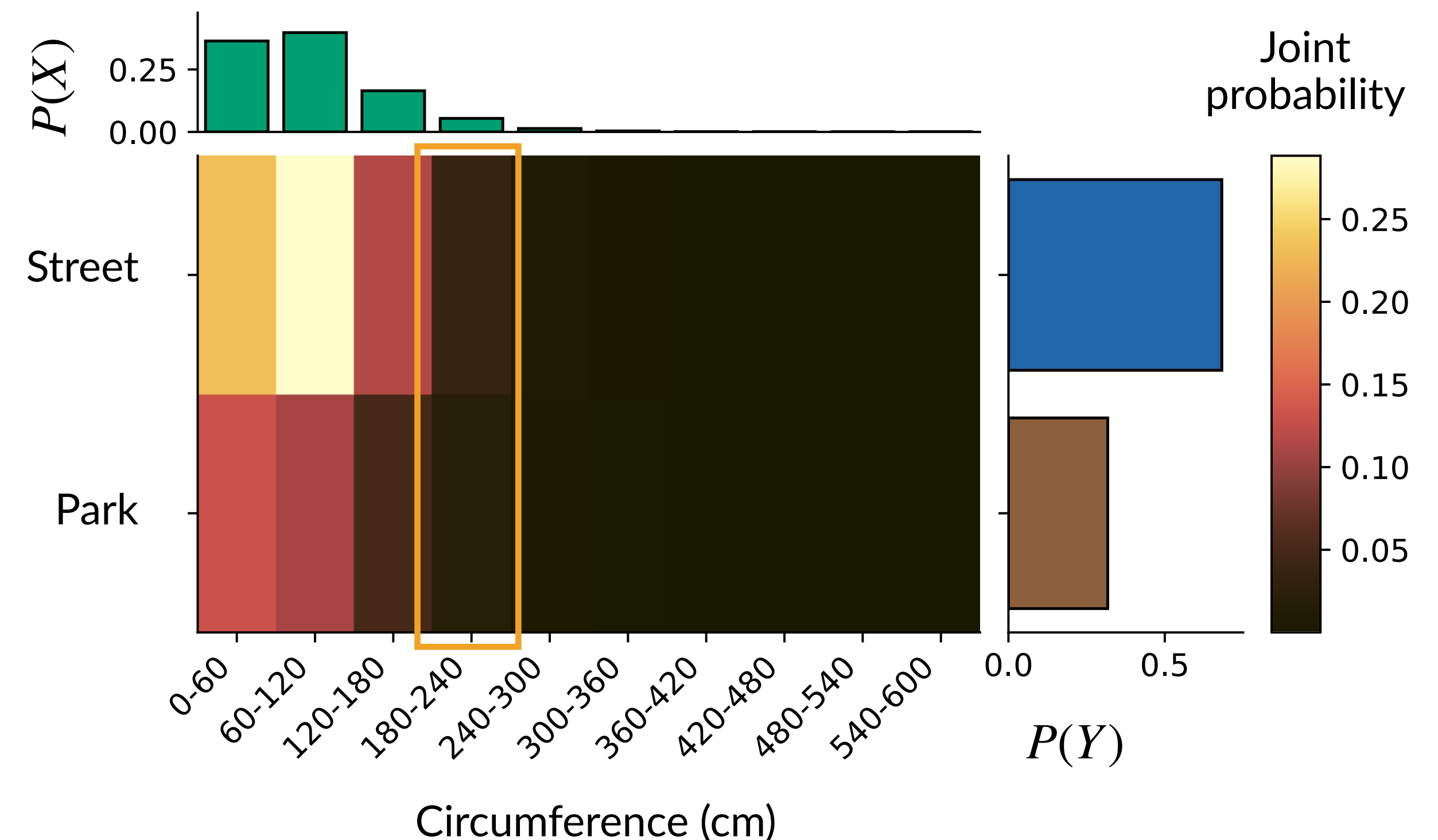
The Grammar of Probability

Bayes' Rule

Q / We've been asking: given the location, what do we expect about size? What about the other way around?
Given the size 180-240cm, what do we expect about location?

Bayes' rule

$$\underset{\substack{\text{Posterior} \\ \text{probability}}}{P(Y|X)} = \frac{\underset{\substack{\text{Likelihood}}}{P(X|Y)} \underset{\substack{\text{Prior} \\ \text{Probability}}}{P(Y)}}{\underset{\substack{\text{Evidence}}}{P(X)}}$$



The Grammar of Probability

A Bayesian View

Q / We've been asking: given the location, what do we expect about size? What about the other way around?
Given the size 180-240cm, what do we expect about location?

Bayes' rule

$$\begin{array}{ccc} & \text{Likelihood} & \text{Prior Probability} \\ & | & | \\ P(Y|X) = & \frac{P(X|Y)P(Y)}{P(X)} \\ | & & | \\ \text{Posterior probability} & & \text{Evidence} \end{array}$$

Prior: what we believe about Y , before seeing any evidence
How likely a tree is to be in a park, before knowing anything about its size.

Likelihood: how likely X is, given a particular value of Y
Among park trees only, how common is a 180–240cm circumference?

Posterior: what we believe about Y , after incorporating what we now know about X .

Once we know a tree is 180–240cm, how likely is it to be in a park?

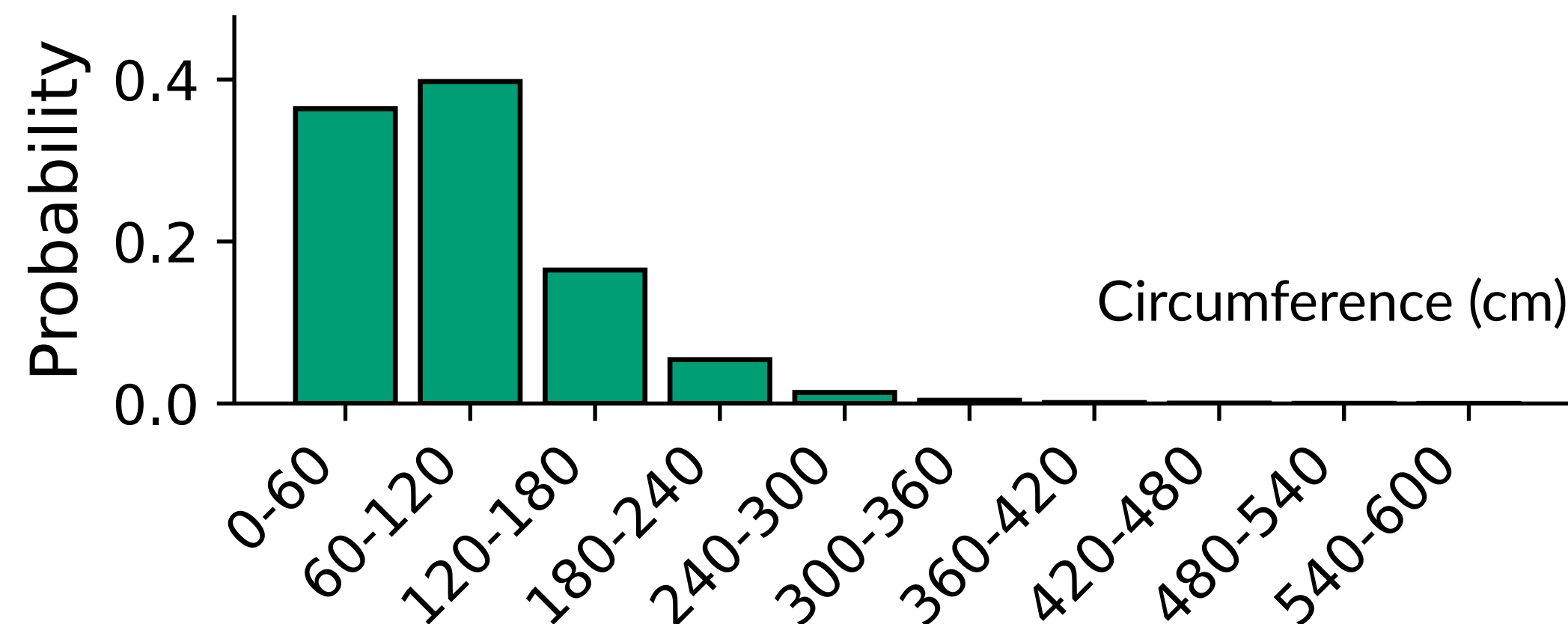
Evidence : the overall probability of X , regardless of Y
How common is a 180–240cm circumference, across all trees, park or street?

Probability distributions

Definition

A **probability distribution** $P_X : \mathcal{X} \rightarrow [0,1]$ is the complete assignment of a probability to every possible value a random variable $X \in \mathcal{X}$ can take

$$P_X(X = x) \geq 0 \quad \forall x \in \mathcal{X}, \quad \sum_{x \in \mathcal{X}} P_X(X = x) = 1$$

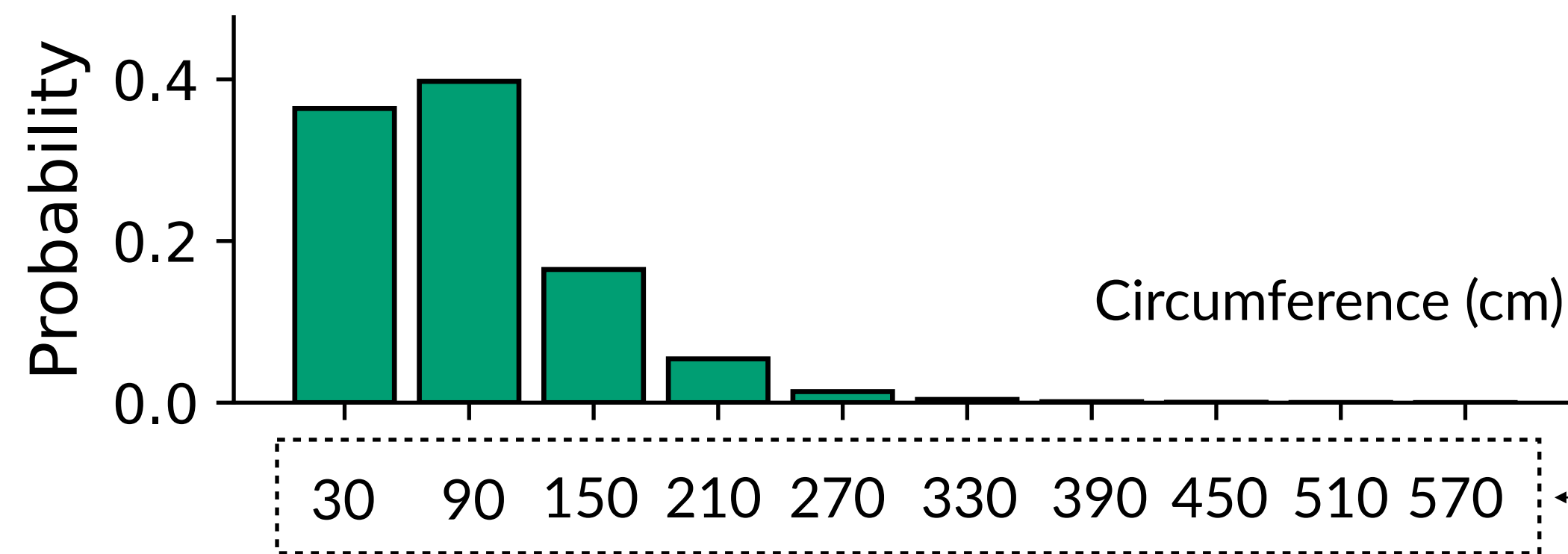


Probability distributions

Expectations

A **probability distribution** $P_X : \mathcal{X} \rightarrow [0,1]$ is the complete assignment of a probability to every possible value a random variable $X \in \mathcal{X}$ can take

$$P_X(X = x) \geq 0 \quad \forall x \in \mathcal{X}, \quad \sum_{x \in \mathcal{X}} P_X(X = x) = 1$$



Q / What is the **average** circumference of a tree in Paris?

$$\text{Mean} / \mathbb{E}_X[X] = \sum_{x \in \mathcal{X}} x P(X = x)$$

The typical value of X —a weighted average, weighting each value by how likely it is

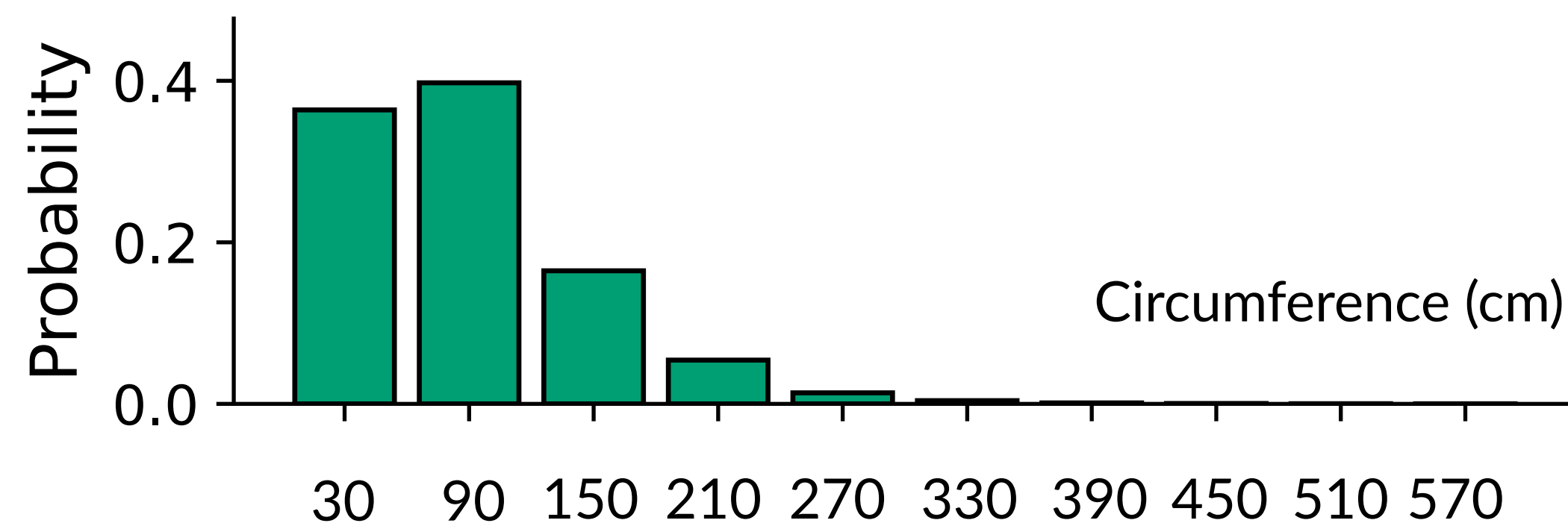
A wrinkle: computing this needs actual numbers to multiply by— but our brackets are labels, like "60–120cm." Each bracket gets its midpoint (90cm) as its representative value—the best guess for a tree we only know falls somewhere in that range.

Probability distributions

Expectations

A **probability distribution** $P_X : \mathcal{X} \rightarrow [0,1]$ is the complete assignment of a probability to every possible value a random variable $X \in \mathcal{X}$ can take

$$P_X(X = x) \geq 0 \quad \forall x \in \mathcal{X}, \quad \sum_{x \in \mathcal{X}} P_X(X = x) = 1$$



Q / What is the **average** circumference of a tree in Paris?

$$\text{Mean} / \mathbb{E}_X[X] = \sum_{x \in \mathcal{X}} x P(X = x)$$

For any function $f : \mathcal{X} \rightarrow \mathbb{R}$, its **expected value** under the distribution $P_X(X)$ is

$$\mathbb{E}_X[f(X)] = \sum_{x \in \mathcal{X}} f(x) P(X = x)$$

For instance, the variance is defined as

$$\text{Variance} / \text{Var}[X] \equiv \mathbb{E}_X[(X - \mathbb{E}_X[X])^2]$$

Probability distributions

Discrete → Continuous

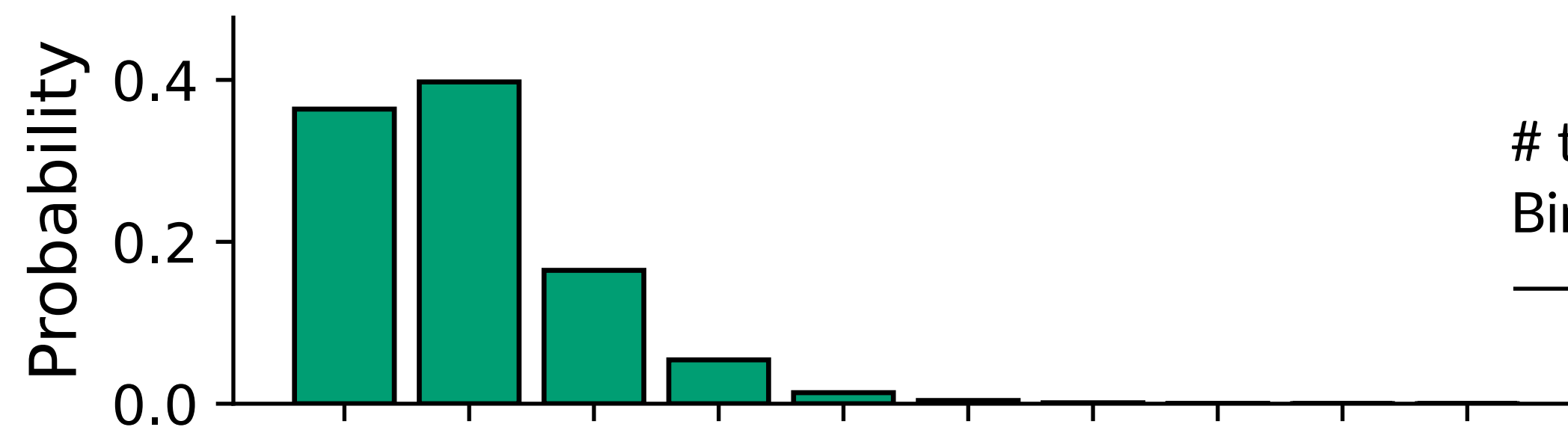
In the limit of infinitely many observations, and infinitely small bins, a histogram becomes a smooth curve—a probability density $p(x) : \mathbb{R} \rightarrow [0, \infty)$

$$P(a \leq X \leq b) = \int_a^b p(x) dx$$

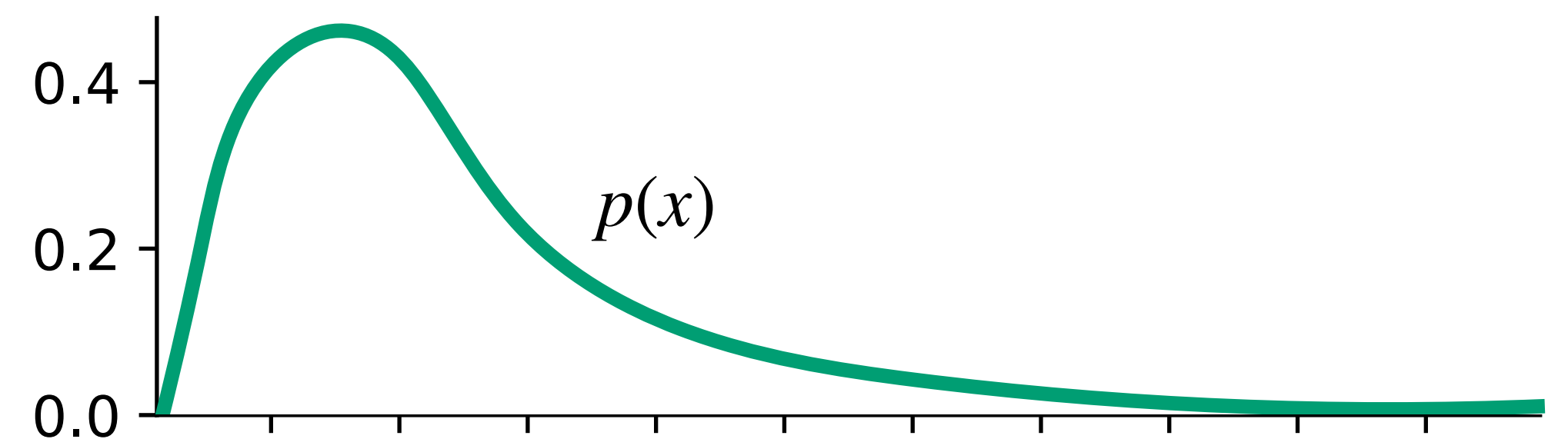
For a continuous variable, the probability of hitting any exact value is zero. $p(x)$ is not itself a probability—only an area under the curve is

$$p(x) \geq 0 \forall x, \quad \int_{-\infty}^{\infty} p(x) = 1$$

But unlike the discrete case $p(x) \leq 1$ is not required!



trees $\rightarrow \infty$
Bin size $\rightarrow 0$



Probability distributions

Translation

Sum & Product rules

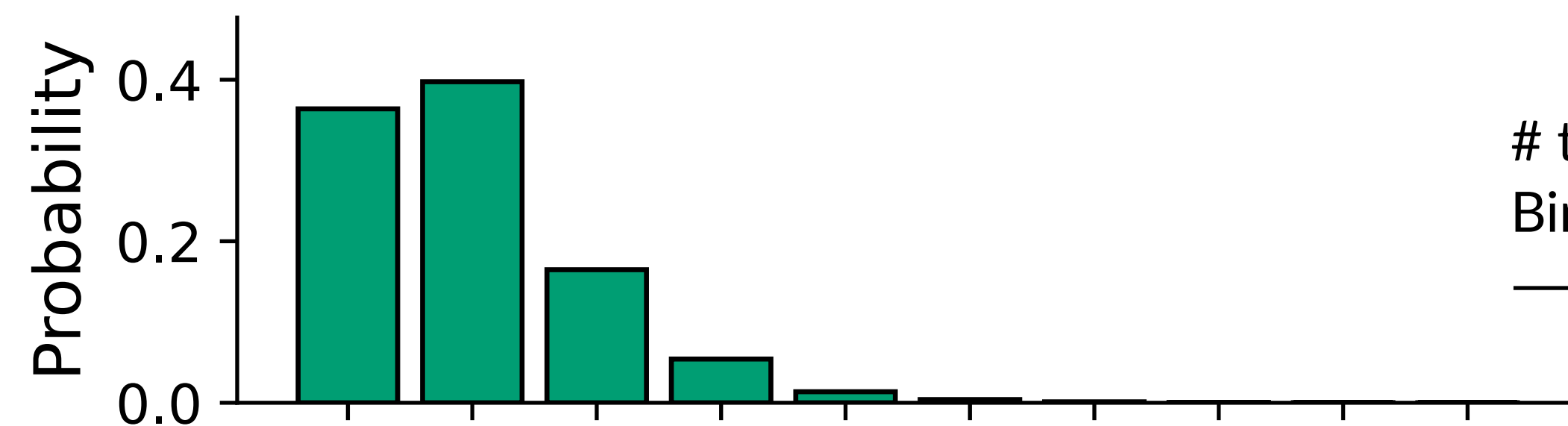
$$p(x) = \int p(x, y) dy$$
$$p(x, y) = p(x | y)p(y) = p(y | x)p(x)$$

Bayes' rule

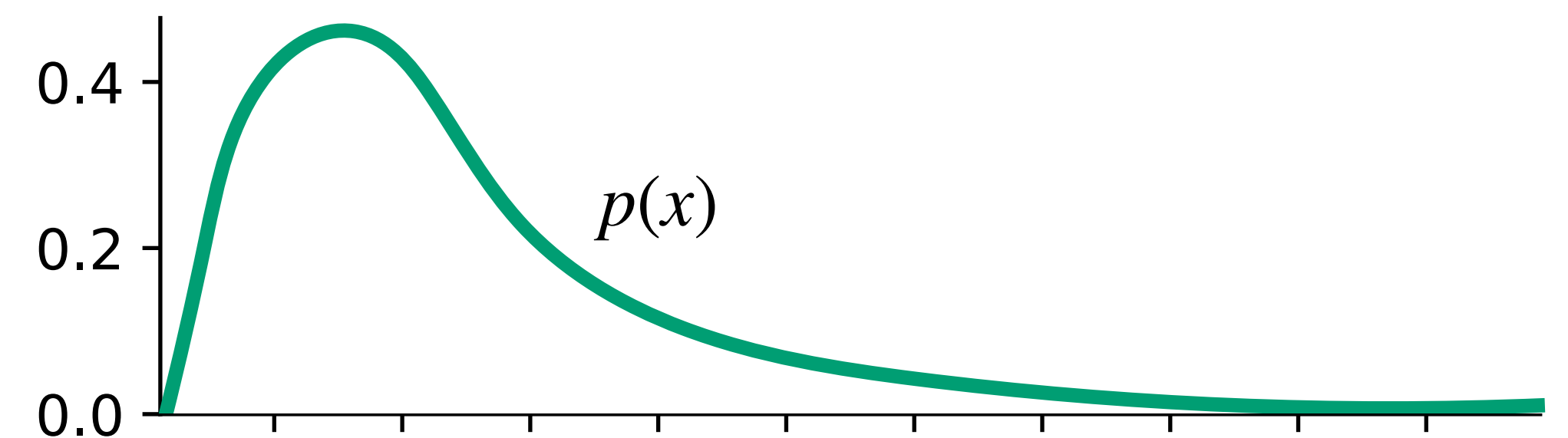
$$p(y | x) = \frac{p(x | y)p(y)}{p(x)}$$

Expectation

$$\mathbb{E}_X[f(X)] = \int f(x)p(x)dx$$



trees $\rightarrow \infty$
Bin size $\rightarrow 0$
 $\longrightarrow$



The Gaussian

The Queen of the Densities

Where averages end up

Sums of many small, independent effects → Gaussian shape, regardless of where they started (Central Limit Theorem)

The most honest distribution, given just mean and variance

Among all distributions with a given μ and σ^2 , the Gaussian is the one that assumes the *least* extra structure —the maximum-entropy choice

It stays Gaussian under almost everything

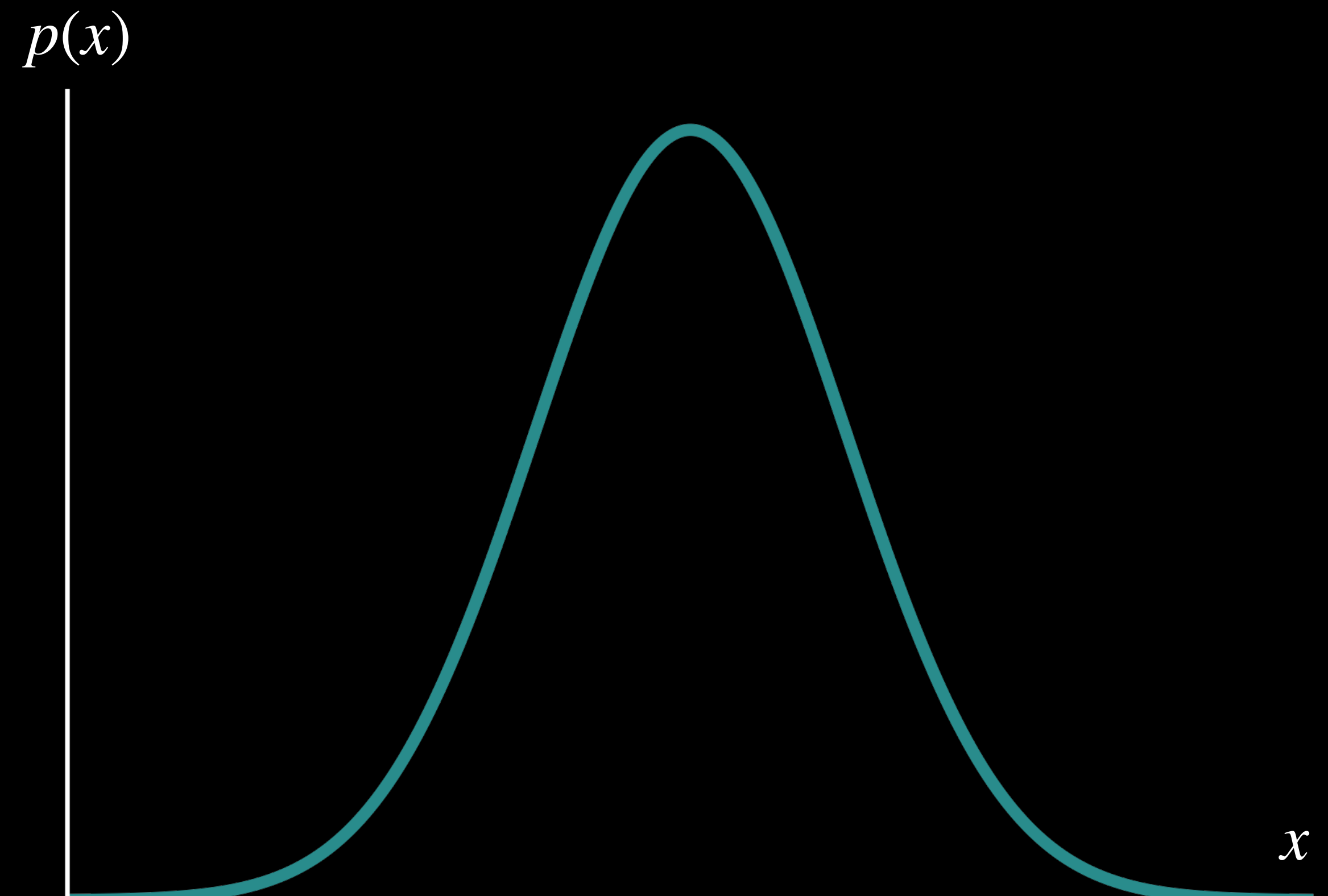
Sums of Gaussians, linear transforms, even conditioning—all Gaussian again. Rare, and extremely convenient

It's the default noise model

Assume Gaussian noise, and least squares becomes maximum likelihood

$$\mathcal{N}(x | \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2}(x - \mu)^2 \right\}$$

$$\mathbb{E}_X[X] = \mu, \quad \text{Var}[X] = \sigma^2$$



High Dimensions

$$p(\mathbf{Y} | \mathbf{X}) = \frac{p(\mathbf{X} | \mathbf{Y}) p(\mathbf{Y})}{p(\mathbf{X})}$$

$$\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu} \in \mathbb{R}^p$$

$$\mathbb{E}[f(\mathbf{X})] = \int_{\mathbb{R}^p} f(\mathbf{x}) p(\mathbf{x}) d\mathbf{x}$$

$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{y}) d\mathbf{y}$$

And most of it generalises well in arbitrary dimensions $\mathbf{X} \in \mathbb{R}^p, \mathbf{Y} \in \mathbb{R}^q$

Proof. The proof is left as an exercise to the reader.

$$\mathbb{E}[\mathbf{f}(\mathbf{X})] = \int \mathbf{f}(\mathbf{x}) p(\mathbf{x}) d\mathbf{x}$$

$$\mathcal{N}(\mathbf{x} | \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{(2\pi)^p |\boldsymbol{\Sigma}|}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\}$$

$$\text{Var}[\mathbf{X}] = \boldsymbol{\Sigma} \in \mathbb{R}^{p \times p}$$

Lecture Report

1 : Introduction



Key concepts:

✓ What is ML

✓ Reminders or Probability

- Joint, marginal, conditional
- Bayes' theorem
- Expectation & Variance
- Densities and Gaussian

Bibliography:

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 **Next week:**
Statistical Learning